

Learn Cybersecurity Fundamentals On The Go

Mammoth Club Official Guide PRO+

Written by Benjamin Stewart • Narrated by John Wayne Holm

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Praise for Mammoth Club

I have completed many tutorials. This one is the most outstanding one that I have seen thus far. It is doubtful that it could be topped. This is a superior tutorial.

Amazing. —Joseph A., Mammoth Club Student

Exactly what I wanted! Just enough BASIC information without being technically overwhelming and intimidating. —Paul V., Mammoth Club Student

This course so far is by far amazing! The instructor is very encouraging and upbeat, and his instructions are very clear. It's an amazing course. —Moiz S., Mammoth Club Student

It's scary to think that by following these instructional videos I can be equipped with the skills to program Python. —Charles E., Mammoth Club Student

I ended up taking it and it was INCREDIBLE. They set great challenges that build off what was taught in the chapter, but don't directly give you the answer. It asks you to extend your knowledge and refer to the right documentation. So good for learning.

—A_Unicycle, Mammoth Club Student

This is AMAZING! I just learned how to code without breaking a sweat, this is really easy and fun! —Shalonda L., Mammoth Club Student

Clear instructions and excellent projects. —Ian F., Mammoth Club Student



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Introduction

Cybersecurity isn't always an easy topic to understand, especially if you're a beginner. You might hear terms like DDoS, spoofing, phishing, malware, encryption, and botnet thrown around. The videos online can be long and confusing, often assuming you already know a vast array of terminology. This can be off-putting, leaving you feeling overwhelmed and confused.

Rest assured, by the end of this book you'll understand what all of those words mean and why they're important. Even if you just pick up a few concepts, it can be very useful in your job or your personal life.

But why? Isn't cybersecurity just for people with degrees who can code or work at Google? After all, it's a highly advanced and rapidly changing field.

The simple answer is that cybersecurity is important to know for anyone who uses a cell phone, laptop, desktop computer, television, or smart device. And let's face it, that's basically everyone.

By the time you're done with this book you're going to know a huge amount more about cybersecurity. You'll know how it started, different types of attacks, how to deal with breaches, how to improve your personal cybersecurity on both your phone and computer, how to make better passwords and much more! If you really find yourself drawn to cybersecurity, at the end of the book we'll even talk about how you can pursue a career in it, what jobs there are, and what the best route to being a cybersecurity expert is.

Cybersecurity is a bit like locking your doors when you leave home. Sure, you might be able to get away with not doing it for a while. But, one day it could go horribly wrong and end up being a painful lesson. Knowing a bit about cybersecurity can help keep you safe. Once you know more you can also advise family, friends, and work colleagues to help keep them safe too. You'll be their resident cybersecurity expert in no time!

Chapter 1: What Is Cybersecurity?

To know and understand exactly what cybersecurity is, let's start with the word itself. As defined by the Oxford Dictionary, the word "cyber" means relating to or characteristic of the culture of computers, information technology, and virtual reality. Simply put, the meaning of cyber is anything that is computer-related. As defined by the Oxford Dictionary, the word "security" refers to the state of being free from danger or threat.

When we put these two together, it's easy to see where the word "cybersecurity" comes from, and what it means. At the most basic level, cybersecurity is the protection of computers and other devices like networks, mobile phones, and servers from threats. These threats primarily take the form of criminals or other unauthorized persons using our private data. This also extends to other malicious types of attacks.

As you'll find out later, not everyone trying to launch a cyberattack actually has a specific and dangerous motive. Some of them are simply bored or trying to learn a new skill. These are often referred to as "script kiddies" because they use existing scripts and software to carry out cyberattacks. These are usually beginner or novice hackers who are stopped more by their own lack of skills than by any moral misgivings. Even though they're not the most experienced or skilled, you still probably don't want them going through your files.

Now that we know what the word "cybersecurity" itself means, we can look at what it means to different people. If you say the word "food" to someone in America or someone in India, different things come to mind. It's the same with cybersecurity; it means different things to individuals, small business owners, big businesses, the government, and IT professionals.

To individuals, cybersecurity is about making sure that their private data and devices are safe. They want to make sure that nobody else can access their email, text messages, private pictures, or financial records. They also want their personal laptops and cell phones to be free from malware, especially the types that slow down their devices or bombard them with unwanted ads. This usually means things like

choosing strong passwords or using a password manager, downloading apps from legitimate sources, and having a password on their cell phones.

For small business owners, cybersecurity isn't all that different from individuals. They need to make sure that all of the computers and point-of-sale devices at their business are secure. The same goes for any work phones, as well as all of the accounting and business data that is stored online. The main difference is that they also need to make sure that all of their employees aren't compromising their cybersecurity.

This means controlling who has access to Wi-Fi networks and with what devices, making sure that employees don't bring personal flash drives to work and plug them into the computers, etc. If the business deals with sensitive information or money it's also important to have a strict technology policy. Employees need to be aware that they can't take selfies at work near sensitive information or do anything else that might compromise the company.

For big businesses cybersecurity can be a much more pressing issue. They need to make sure that they have specialists in charge of the department. All work computers need certain software and employees need to be clear about what they can and can't do. It isn't just enough to tell them this, if you have thousands of employees some of them are bound to mess up. For this reason, many big companies use specialized software that encrypts any flash-drive put into a computer and protects other documents. It's important to remember that it isn't just their own data that they need to protect. Companies like Meta (the owner of Facebook, Instagram, and WhatsApp) are looking after data for billions of people worldwide.

The tiniest leak or exploit in their systems could mean massive scandals as well as lawsuits. Big companies are also a much bigger target for cyber attackers. The more people a company interacts with, the higher chance it has of coming across those with the means and desire to commit malicious attacks. Depending on the type of business the risk can also be much greater. Accounting and financial firms will always have people trying to access their confidential records or edit their data. Likewise, banks are a massive target as every would-be hacker is looking for a way to get rich quickly. Big companies need to take all of these risks into account and make sure that they've invested enough into their cybersecurity.



To governments, cybersecurity is all about laws and procedures. Data needs to be classified in different ways typically public data, private data, confidential data, internal data, and restricted data. Each of these data classifications needs its own set of laws, policies, and procedures. As the digital world is constantly evolving, governments are scrambling to keep up with legislature. Pictures that you store privately on your cell phone or that you post on Instagram or Facebook don't have the same rights protecting them. It's up to policy-makers in the government to decide what rights those pictures should have and based on what criteria.

To IT professionals, cybersecurity is a way of life. It's how they make money, feed their families, and make a name for themselves. Some of them are responsible for protecting their company or many companies from potential threats and attackers. Others are responsible for developing the software used to protect against hackers, viruses, and other cyber-threats. They don't just need to keep up with the attackers, they need to be one step ahead at all times. This means thinking like a hacker, and often being one.

These individuals are known as "ethical hackers" and companies pay them to find exploits in their systems. Many companies have adopted the stance that it's better to pay someone ethical to find these problems than wait for someone unethical to try and exploit them. Once they've found these, they usually help the company by advising them on ways to fix the problems and improve their systems.

I'm sure you can see that cybersecurity means a wide range of things to a wide range of people. But, there's one common theme. It's important to all of them in one way or another. It protects people's privacy so that they don't have to have everything they do or care about exposed in public. It protects countries' classified information and important infrastructure so that important sensitive information isn't leaked or destroyed.

For businesses it stops attacks that would stop them from being able to operate and provide goods and services to the public. It's only because of cybersecurity that worldwide connection and communication is possible. Without it, connections would constantly be severed and many people wouldn't communicate for fear of their conversations being leaked.

Cybersecurity is crucial for maintaining the security, privacy, and operability of a variety of digital systems. It is also something that's constantly changing and evolving. 40 years ago, social media didn't exist, and if you mentioned posting pictures online people would have looked at you like you were crazy. Cybersecurity didn't exist for smartphones because phones weren't smart. They couldn't take pictures and even if they could have, they would never have been able to upload them to the internet.

This means that it's an already large field that's only going to grow with importance as time goes on. As people use more smart watches, fridges, vacuum cleaners, light bulbs, speakers, air conditioners, and even beds, there are more vulnerabilities in every home.

They also put more and more data about themselves online or on digital devices that could potentially be attacked. There are more places that attackers can look to find a way in. This already important field is only going to further explode as homes and workplaces are flooded with all kinds of new smart devices, every one of them a potential security incident just waiting to happen.

Chapter 2: Passive Cyber Attacks

Cyberattacks can broadly be split into two categories. The first type of attack that we'll be looking at are passive attacks. The simplest way of thinking about this is that the attacker doesn't actually *do* anything to you. They don't send you suspicious emails, stop your phone from working, or delete all of your files.

Instead, they simply watch and wait. The term used for it is *eavesdropping*, like when a person secretly listens in on a conversation that they were never a part of. These attackers listen in on as much of your communication as possible, using it to discover more information about you. They don't change any of the data though, they just observe it. This makes these types of attacks very difficult to detect as your whole system will still function normally.

Most of the software used for these types of attacks falls under "spyware". This is because it "spies" on the user, taking their information so that people or companies can use it for their own purposes. Spyware is often bundled with other useful software or masks itself as a useful program. This is because it's much easier to find vulnerabilities within a user than within a system.

A user will click through the terms and agreements of a program installation, without noticing that one of the checkboxes is for installing an entirely different program altogether. Once they have consented to the installation being done, the spyware will be installed. Many users don't even realize that this has happened until their computer starts slowing down.

But how do they actually track you? What information are extracting?

There are a variety of ways they do this, the most commonly known one is keylogging. There is also wiretapping, fiber tapping, port scanning, and idle scanning, as well as other less commonly known types.

Keystroke logging, commonly referred to as keylogging, is recording every key that is hit on a keyboard. The person using the keyboard is never aware that this happened, however the data can later be sorted and used. These keyloggers can either be a physical device that is plugged into a computer or an incredibly small piece of

software that can easily go undetected. Windows 10 even has a keylogger built in! Microsoft says that it's there for general typing improvement.

This goes to show just how easy a keylogger is to develop and implement. The hardest part for potential attackers is to get the keylogger onto a system without being noticed. The second hardest part is for them to get the information from the keylogger off of the system without accidentally exposing their IP or email address.

It's important to note that keylogging isn't actually illegal, many employers use it as a way of monitoring their employees at work. It can also be used to help with tech support or to monitor children's devices. The problem is, most keyloggers are used for nefarious purposes. If you log everything someone types, it isn't too much trouble to figure out their email password, banking login details, and all kinds of other confidential information.

Once a hacker has this information it's incredibly easy for them to use it to steal money or personal details like an ID or social security number. These details can be used to impersonate someone or to make other types of attacks like phishing more believable.

You may already have heard of telephone tapping or wiretapping from movies or series. In movies you'll often see a government agency put a wiretap on someone's phone, this is also known as "lawful interception". In most countries it is very difficult for government agents to get authorization to do this and unless you're involved in something highly illegal it's very unlikely that they would be authorized to wiretap your phone.

The name "wiretapping" originates from a time when everyone used physical telephones and it was an electrical tap on a physical phone wire. Hence the name "wiretap". While physical wiretapping does still exist, cellphones are much more widely used. For this reason, most wiretapping nowadays is done using software.

The actual infrastructure used for cellular towers and "normal" phone calls is Signaling System Number 7 and it was developed in the 1970s. As you can imagine, such old technology has some pretty large security vulnerabilities, and the system is well known for its lack of security. Unlike modern apps like WhatsApp, Signal, and Telegram, Signaling System Number 7 is not end-to-end encrypted. This means that



anyone with access to the exchange can eavesdrop on phone calls and texts as well as access your phone's location data.

Even without being an employee at an exchange, hackers have found ways to passively intercept people's mobile calls. This is because many mobile companies, especially in less wealthy countries, still use outdated 2G networks. The equipment needed to do this can be built from easily available parts. In addition to this, free decryption software available on the internet can be used to hack into these 2G systems and monitor both text messages and phone calls.

In order to prevent these types of attacks, it's always best to use end-to-end encrypted apps for calls and messages, especially if you're dealing with sensitive information.

If you're using one of these apps then the only way for people to "wiretap" your phone is by hacking your actual cellphone and getting malware onto it. This malware can then log what you're typing, like the keyloggers we spoke about previously, or listen in via your microphone. The malware that listens in via your microphone has become more advanced in recent years, and it can even listen to you while you aren't making a phone call or voice note.

While it's not impossible for this to happen to you, it's much less likely as actively hacking someone's phone to get malware onto it can be a lot of time and effort. For this reason, it most commonly happens to high level politicians and celebrities, people who are naturally targets.

Another common method of passive cyberattack is fiber tapping. The general idea is not that different from wiretapping used in phones. Not the more modern kind where hackers put malware onto your cellphone, but rather the older kind where they physically monitor your line with an electronic device.

Fiber optic internet lines use thin strands of plastic or glass to transmit signals as pulses of light. Each pulse of light represents either a 1 or a 0. If you've ever heard of binary code before, you might see where this is going. This means that all of the information transmitted through fiber lines is sent as ones and zeros. As light travels incredibly fast, so do the ones and zeros, allowing for much faster connection speeds than traditional copper internet cables.

In order to save costs, when internet companies lay fiber to homes, they use devices called “beam splitters”. Without getting too into detail on the inner workings of fiber, the devices split the light, allowing for multiple users to use the same line. When a fiber company sends a technician to check if the line is working, they simply put a bend in the cable and check to see if light is being transmitted.

Similar methods can be used to tap fiber cables for surveillance purposes. If a person can put a tap on the line to receive the light signals, they can figure out what information is being sent. The problem with using this method is linked to the beam splitters that we spoke about. As multiple home users share one line, when attempting to eavesdrop on one person, you can easily receive the information for all of them. Sorting this information can be a massive task, especially considering how many different devices and applications even a single home can have.

It is also possible that people will notice that the fiber line has been tapped, as a result of the increased attenuation. To put it in the simplest way possible, the “power” of the light being sent will decrease at the point of the tap. Fiber companies monitor this to make sure that there are no physical issues with their lines. If they notice significantly higher attenuation or loss of power in one place, they will probably send someone to investigate the issue and try to pinpoint where exactly it is.

In addition to this, once hackers have the information from their fiber tapping, they have to be able to figure out how to read it. If you’re using an app with end-to-end encryption like Signal, Telegram, and WhatsApp that we spoke about previously, this makes it harder for them. Once they have the information from your calls or messages, they still can’t figure out exactly what it says as they don’t have the encryption key.

Port scanning is a type of passive attack that is designed to check a server or a host for open ports. You can think of a port like a virtual door that allows for communication between different software services and/or applications.

Sometimes IT professionals use these scans to make sure that there are no vulnerabilities on their systems. Other times, would-be attackers use it to find vulnerabilities on certain networks. It is sort of like checking a building to see which doors are open. You could be doing this because you’re a security guard and you



want to make sure everybody is safe and remembers to lock them. You could also be doing it because you want to take your neighbor's laptop while they're not looking.

A port scan will allow someone to see which ports are open and which are closed. For example, a certain type of email might be using port 110 and web traffic might be using port 80. As you can see, a port scan in itself is not inherently malicious, it's just checking whether or not ports are being used.

The problem comes in once the attackers have determined which ports you're using. They can use this to determine their best point of entry. They might see that your email uses port 110 and then use this to try and gain unauthorized email access or try to attack your email client.

The easiest and most efficient way of mitigating these attacks is to use a firewall. Many operating systems like Windows 10 come with a built-in firewall. It's important to make sure that it's active so that no unauthorized users or programs can gain access to your system.

The last type of cyberattack that we will look at is idle scanning. Idle scanning is a way of checking what apps are open on any given computer. Essentially, they use something called "packet spoofing" to impersonate another computer. They then send information to your computer in the form of a SYN packet (a SYNchronization packet of data) as if it were from someone else.

Depending on how your computer responds they can tell whether or not the port is open, and if you're using a firewall. For example, if there is no response at all, it means that the system they're targeting is using a firewall.

If the system responds with an RST, a reset packet, then it means that the system is rejecting the connection. This would be because the port is closed. If the port were open, the computer would respond with a SYN-ACK packet (a SYNchronisation Acknowledge packet of data).

Because your computer doesn't think these packets are coming from the attackers, it can't be traced back to them. The computer is then never aware that it's interacting with the attackers, which allows them to do this whilst remaining completely anonymous and invisible to the target.

A bit later on we'll cover more in-depth ways to prevent these kinds of attacks and keep your data safe.

Chapter 3: Active Cyber Attacks

Now that you've learnt more about the passive ways in which attackers monitor you and attempt to obtain your information, it's time to look at some of the more active ways in which it happens. These attacks don't just look to collect data and spy on you like some of the passive attacks. Their intent is to cause harm by compromising computer systems, networks, and digital infrastructure.

These attacks only go unnoticed if they fail and the hackers can't manage to pull off the disruption that they intended to. If they succeed, you can be sure people are going to notice. Active attacks take websites down, hold people's files and computers hostage for money, and disrupt general operations.

A distributed denial of service attack, commonly referred to as a DDoS attack or DDoSing, is the first active type of cyber attack that we will be looking at. The goal of this type of attack is to make a service unavailable to its intended users. This is done by sending a flood of traffic to a specific host or resource.

If this was done using only one computer it would be easy for the host to simply block the source of the unwanted traffic. The first D in DDoS stands for distributed, and this refers to the fact that the flood of traffic originates from many different sources. Because the flood of unwanted traffic is coming from so many different places, it's very difficult to block all of them. This often means that the only viable way to counteract the attack is to increase the capacity of a website or service so that it can handle the extra traffic. You'll see how this can sometimes be used against the target a little later.

You can imagine this attack like a group of people deciding to flood a grocery store after work because they don't like the owner. None of them intend to buy anything, they just want to get in the way of normal customers. If it was just one customer causing a scene the owner could simply ban them from buying at the store.

If it was hundreds of people, it could be difficult for the owner to figure out which ones were real customers and he wouldn't want to accidentally ban the wrong person. This will very quickly disrupt the business's trade and cause them to lose

money. If it continues like this, most customers will simply find somewhere else to shop.

DDoS attacks work in the same way. The goal isn't to permanently damage something, it's just to stop it from working for long enough to cause a disruption to users. Some types of DDoS attacks known as yo-yo attacks aren't constant. The attackers allow the target time to scale up their resources in order to handle the increase in traffic.

Once they've done this, they stop the attack. The target then realizes that they have more resources than they need and scales down. As soon as this happens, the attack starts up all over again.

Distributed denial of service attacks are sometimes used for blackmail, but they're also a commonly used tool of hacktivism. These are people who use hacking to push a specific social or political agenda. They often work in teams with like-minded individuals, giving them access to more resources and making the attacks more effective.

Another category of active attacks is spoofing. During these attacks, the attackers disguise themselves as something or someone else. This is done by falsifying certain data and the goal is to deceive the target and gain an advantage by doing so.

Imagine that you receive a message from your spouse's number saying to leave the door unlocked when you go to work because they lost their key. If it were another random number asking you to leave your door unlocked, you would know that it's a trick. But, if you believe it to be your spouse, you're much more susceptible to their suggestions.

You can think of spoofing attacks as tricks. For example, attackers will often falsify an email address. The goal is to trick users into thinking that an email they received is from a legitimate source. This will make them more inclined to click a link in the email, listen to the instructions, or download the attachments.

Sometimes all they want is for the person to disclose personal information in response to the mail. This information can then be used for spoofing other individuals, for blackmail, or for gaining answers to their security questions. SMS

spoofing works in very much the same way and it's common for it to be used for social engineering or phishing attacks.

An IP address is a unique address that is used to identify a device connected to the internet or a local network. For example, the IP address for your router might be 192.168.0.1. You can think of this as an ID number for computers. When attackers make use of IP spoofing, they change a packet of data to make it look like it's coming from a source that your computer trusts. This can help them to gain access to systems, launch DDoS attacks from IP addresses that your network trusts, and even spread malware.

Another type of spoofing is website spoofing. The attackers impersonate a common site, maybe a banking site or a popular online retailer. Because they can't use exactly the same web address, there will be slight differences. They will often use a common misspelling of a word or a variation that looks similar.

They might spell PayPal with an "E" instead of an "A" or something similar. Once users enter their login details, these will be stored in a database. The attackers can then use these details to access the legitimate site themselves.

These attacks are often used in conjunction with one another. For example, an email will be sent to a target from a fake email account, pretending to be a legitimate organization like their bank. It will inform the user that there is a problem with their account and they need to login to correct the problem.

The spoof email will then provide a link to a website that looks almost like their normal banking link but is slightly different. This allows the attackers to target specific individuals instead of just waiting and hoping that somebody misspells the name of a website that they commonly use.

Attacks that use different approaches like this are commonly known as mixed threat attacks. The attackers need to be competent at a variety of different skills to pull off a mixed threat attack but they can be incredibly effective. Like in our previous example, it might start with a fake email. However, instead of looking for login details, the email could prompt users to download something.

This something could be ransomware. You might be wondering, what is ransomware?

It's just like any other kind of ransom, only with a computer. I'm sure you've heard of kidnappers stealing somebody's child and demanding a ransom in order to let them go. Ransomware accesses a computer's files and then encrypts them so that the person cannot see or use any of their files. After this, the computer will usually display a ransom note, informing the person how much money they need to pay and to where. This will usually be accompanied by a timer that ticks down. When the timer reaches zero it will usually either delete the victim's files or demand more money.

The goal is for the timer to be long enough that the victim has time to pay the money, but also short enough that the victim doesn't have time to consult a cybersecurity professional for help with the issue. The ransomware will also often try to spread itself to other computers on the network.

Small businesses and individuals with lots of personal information on their computers can be most at risk. This is because they don't have the necessary software to protect themselves, but they also have a lot to lose. To someone with years of pictures on their computer that haven't been backed up, the thought of losing all of these can be soul crushing.

Another popular attack is the man-in-the-middle attack. To understand this, imagine that you have a friend in another country who you message regularly. Their name is Jeff. When you send Jeff a message you assume that he is reading and replying to it. What is actually happening, is that the message is going to a third party, a cyber attacker named Steve. Steve takes your message, reads it, makes a few changes, and then sends it on to Jeff. Whenever Jeff sends you a message, the same thing happens.

Both of you are convinced that you are communicating directly with each other. In reality, you're both talking to Steve and he chooses what you see. This is different to a passive eavesdropping attack like a wiretap because the attacker can actively change the information that both parties are receiving.

Most modern communication apps have safeguards to prevent this from happening. There are however still ways that these attacks can work. For example, if you use an unencrypted public Wi-Fi network. An attacker who is also connected to this network could insert themselves as a man-in-the-middle.

There are many more types of cyberattacks, both active and passive, but the last one that we will take a look at is zero-day exploits. A zero-day exploit is a vulnerability that is found in software before the developer of the software is aware that it exists. These types of software exploits are one of the main reasons why applications have to release frequent security patches after they are released.

Developers try their best to make sure that their software is secure but it's always going to be difficult to think of every possibility. Traditionally antivirus programs relied on pattern detection to identify malware. The antivirus then picks up anything that matches the database of malicious code and classifies it as a threat. Depending on the program and the user it will then be quarantined, deleted, or they will simply be notified. The problem with this type of detection is that zero-day exploits don't match anything in the database.

Because they are new exploits for new programs, the antivirus program doesn't know whether they're malicious or just any other piece of code. Because of this they can be incredibly difficult to detect before it's too late. They can also affect a large amount of people, as it's not a problem with one specific user or individual computer, it's a problem with the code that has been downloaded by thousands or even millions of people.

Chapter 4: Who Are the Attackers and How Do They Choose Their Targets?

The first important thing to note is that not all attackers are alike. We'll start with the "good" hackers and slowly make our way down to the worst types.

These good hackers are known as "white hat hackers". This group is made up of cybersecurity professionals who are paid for their hacking services by companies. They're using their skills for constructive purposes like enhancing security, preventing cyber threats, and protecting systems.

They're commonly involved in fields like penetration testing, which we'll talk about a little bit more in the next chapter. Overall, they just want to use their skills to earn a living and make sure that they're keeping people safe.

Next are the grey hat hackers. These hackers often don't have the proper authorization for the activities they're conducting. Despite this, they do so without malicious intent. They often inform the affected parties of vulnerabilities that they have found after they're done with their testing. Even if they don't do this, they won't actively exploit the vulnerabilities. While what they're doing isn't strictly legal, it isn't malicious either.

They're simply trying to learn more about cybersecurity or are sometimes motivated by a desire to help companies. They're often teenagers or university students learning about the field who want to see if they can apply it in real life. After all, who wouldn't be curious to see if their skills are actually useful in the real world?

As you might have been able to guess from the previous two types, next come black hat hackers. These are individuals who hack for malicious purposes. Any vulnerabilities that they find will be exploited. They want to steal data, disrupt services, and carry out other illegal activities. Equipped with loose morals and a desire for financial gain, they're out to take what they can and give nothing back.

Even within this category, there are different types of black hat hackers. The ones that could be seen as the “least bad” are the hacktivists. This is because, despite their malicious intent and illegal activities, they’re doing it for a cause that they believe in.

They might be disrupting the business of a company that they believe exploits workers, or hacking to bring attention to a specific political or social cause. While what they’re doing isn’t legal, they’re unlikely to sell the user’s data to the highest bidders. The problem is, when people are acting outside the law for their own agendas, things can go wrong quickly.

Even if the hacktivists have good intentions, the people they get involved with to carry off the attacks often aren’t. In addition to this many innocent users can be harmed as the services they rely on stop working or their data is leaked along with the company’s data.

Next, come script kiddies. Much like grey hat hackers they’re often motivated by curiosity. They don’t however have the same moral compass or desire to help the companies that they find vulnerabilities in. Instead, they’re using tools developed by other more experienced hackers to carry out their own attacks.

They lack a deeper understanding of the script that they use as well as the underlying technology. They are often in it for the thrill of it, looking to prove their abilities and gain bragging rights without putting in too much work.

There are also state-sponsored hackers. How ethical you believe their activities to be largely depends on your personal beliefs and the country that you live in. They’re paid by governments to conduct cyber espionage and other activities that advance the interests of the people in control of their country. This could include activities like attempting to influence other countries’ elections and committing cyber warfare.

Lastly, come cyber criminals. These are considered the worst of the worst when it comes to hackers. They’re the ones most likely to be carrying out illegal activities like ransomware attacks, identity theft, and credit card fraud. Their motivations are purely financial and they make a living off of stealing from others. The simplest way to think of them is as cyber thieves, stealing anything and everything that they believe could make a profit.

This is a large group of hackers as research suggests that 86% of all data breaches are financially motivated. This means that most hackers are choosing their targets simply based on who they believe they can get the most money from. As mentioned earlier, this is why many financial services companies like banks, and other larger businesses are victims.

But a lot of the time the hackers aren't actually going directly for the money. They're trying to find sensitive information on users that they can then sell on the dark web for money. The more customers a business has, and the more sensitive information it keeps on them, the better of a target it makes for most hackers. Money is easier to trace and getting it from the victim into their bank account without being found out can be a difficult process.

For this reason, they would often much rather take valuable information than the same amount of money from a victim. The money that they can get from selling the information will be much harder to trace as the authorities can't easily find who the data was sold to or what money was used. These cyber thieves are all about getting as much as they can as easily as they can.

The second main factor that they look for is vulnerability. Most hackers aren't going to target a very difficult target like a cybersecurity company. Instead, they're going to look for vulnerable targets. These companies might be using outdated software or hardware as well as have no dedicated cybersecurity professionals. Often, they don't educate their staff properly on recognizing cyberthreats and avoiding them.

Even within a company, they'll look for the least knowledgeable and most gullible targets possible. Simply by reading this book and understanding the methods they use, you've already made yourself less of a target for potential attackers. They don't want to target people who can identify what they're doing and report it, they want to target people who know as little about cybersecurity as possible.

Another main factor that the hackers look for is disruption. They want to find an area where they can cause as much chaos as possible. Going for a business in financial services doesn't just disrupt the business itself, it also disrupts all of their customers. Customers who are understandably concerned about where their money is and that it's being taken care of.



Any breach to a financial services business is considered a major one for their customers because it's an area of their businesses and lives that is extremely important. This is also why they go for big companies like Facebook and TikTok.

Despite the fact that they're not very vulnerable and they have plenty of cybersecurity professionals, they still make a juicy target. This is because of the sheer amount of data available, which ties to the first point of financial gain, and the amount of disruption this can achieve. Imagine how big of a news story it would be if Facebook was hacked and everybody's personal data was leaked tomorrow.

These are the obvious things that hackers look for but it's important to understand that they're all different. You or your business could be a target simply because you made the wrong person angry. Or because they live near you and they saw your business's sign on the way home. Perhaps the type of hackers who are interested in you aren't after money, rather they form the other 14% who are hacktivists or just woke up and decided to break something.

This is why it's important to always follow the correct cybersecurity protocols and make sure that you aren't an easy target. Later on in the book we'll cover how to improve your personal cybersecurity on your phone, computer, and even in your business.

Chapter 5: How Are These Attacks Carried Out?

How do these hackers actually find and exploit their targets? How are these attacks actually carried out? We've already covered various common types of attacks as well as who the hackers are but there's a big jump between knowing the attacks exist and actually being able to implement the techniques required. How do the attackers make sure that the malicious software goes where they want it to be? How do they make sure they're not accidentally going for the wrong target?

Once they've found their target, preferably someone vulnerable who they can make money from, where are they getting the code from? How do people like the "script kiddies" that we spoke about earlier, who lack in-depth programming knowledge, find the code to use? Obviously, Google isn't publishing useful results for, "Please help me hack my neighbor's Wi-Fi, I want free internet."

Like with any other software development, the most experienced hackers know how to code their own tools, in this case, malware. This is essential to tailor the software for individual targets and exploit new code like zero-day exploits. This does however require an advanced knowledge of programming and cybersecurity specifically.

So, what about the so called "script kiddies" who don't have the necessary experience or skills to code their own malware from scratch. Where do these people with limited knowledge get software that they can use for illegal purposes?

The first thing to understand is that not everything on the internet can be found using your web browser. You can think of the internet like a mango. On the outside of the mango, the skin, we have the Clearnet. The Clearnet is the normal internet that you can access anywhere from any browser. Naturally, this excludes countries with very strict internet policies that block common sites. Some common examples of things on the Clearnet include Facebook, YouTube, and TikTok. These are sites that most people know about and use regularly. Like the outside of a mango, everyone can see it and it doesn't take much work to get there.



But the skin is actually a very small part of a mango. There's much more underneath the skin that you can't see unless you cut it open. This is the deep web. This part of the internet isn't indexed. This means that no amount of searching on a regular browser will get these parts of the web to pop up.

However, if you know the URL for what you're looking for, you can often still access the sites. A lot of the stuff that's here isn't actually all that interesting. A lot of it is just data storage, the code that actually makes the websites you see run, and other mundane stuff. It isn't indexed not because it's controversial or illegal, but rather because nobody would want to look at it. It might also be private information that isn't bad but people also don't want just anyone looking at it.

At the center of the mango you have the seed, and let's face it, if you've gotten here you've probably gone far enough. There's nothing good inside it, it's bitter, and you'd be better off looking somewhere else for a tasty snack. Nevertheless, it exists. This part of the metaphorical mango is the dark web. Like other large sections of the deep web it forms a part of, you can't access it from a normal browser like Google, Edge, or Firefox.

In order to access this part of the web you need a special browser like Tor. This browser can read the .onion URLs that the dark web uses and take you to the sites. On it, there are all kinds of illegal things, including marketplaces selling drugs and even people. It's a place that can only be accessed by those who are good with computers. And most people who are good with computers and stay on the right side of the law have no reason to access it.

As you can imagine, a place that congregates people who are good with computers and of questionable moral standing is going to be a place where lots of hackers, script kiddies, and other unsavory individuals congregate. This is the place where some potential attackers find the guides and software that they need to perform cyber attacks. There are also forums dedicated to these kinds of discussions.

Despite this, most of the information needed by hackers can actually be found on the Clearnet. It's just about how you look for it. Penetration testing, often shortened to pen testing, is an exercise where cybersecurity experts try to find vulnerabilities in a given system. And, because the owner of the software gave them permission to do it, this is legal.

I'm sure you can see where this is going. The tools used to legally pen test a computer system or network are the exact same ones needed to illegally access a computer system. Take a tool like Wireshark for example. Wireshark is a popular open-source tool that can be used to capture and inspect data from a computer network.

Cybersecurity professionals often use it to do security assessments and troubleshooting. It can be used to monitor the traffic on the network, decode packets of binary data into formats that people can read, and troubleshoot issues with their network.

A malicious attacker could take Wireshark and use it to perform various unauthorized activities. They could capture people's packets and inspect them to see what they're doing. They could intercept the user's network traffic and change it, like the man-in-the-middle attacks we already spoke about. Or perhaps they could examine user's IP addresses and use this for network spoofing in the future. They could even be inspecting the network for potential vulnerabilities to use later when launching a DDoS attack.

Wireshark is only one of many programs. There are plenty of other types, including password cracking tools like John the Ripper and Hashcat. Once attackers have captured a packet of data, for example a request from a phone to connect to a Wi-Fi network, they can attempt to crack it.

These programs can attempt to crack passwords in a variety of ways. There is the brute force method. This tries all of the possible combinations of numbers, letters, and symbols until the password is found. As you can imagine this takes a very long time.

For this reason, it's more common to use something like a dictionary attack or a rule-based attack. As it's much more likely that somebody will make their password a real word, dictionary attacks use lists of known words to guess passwords. A rule-based attack accounts for some of the changes people make to dictionary words. For example, using a dollar sign instead of an S when making their passwords.

These methods can be effective but they can also be time consuming and yield no results. Because of this it's often easier for attackers to use social engineering methods like spoof emails, pretending to be tech support, and making fun online



quizzes that are actually asking users for the answers to their security questions. There are completely free and open-source programs like Gophish that are used by penetration testers and hackers alike.

Gophish allows users to design phishing emails with realistic looking content that can be used to obtain people's personal information. The program helps to schedule phishing campaigns, monitor people's responses to the emails, and see who clicked on them. It's a very useful way for cybersecurity professionals to make sure that employees are being safe and adhering to safety protocols.

Any employees who do click on the fake emails can be sent for training to ensure that it doesn't happen again and that they learn what to look for in fake emails. It's also a very useful way for potential scammers to design their campaigns and make legitimate looking phishing emails.

As you can see, the difference between ethical penetration testing and illegal hacking all comes down to permission and intent. It's the same skillset used for both activities. It's a bit like if you're good at chemistry. You could choose to study a degree in chemistry and use your skills to teach children or university students.

Perhaps you could use it to work on medications that sick people need. Alternatively, you could use those very same skills to make illegal drugs. The difference between ethical penetration testing and illegal hacking isn't in the methods or skillsets, it's in what they choose to do with the vulnerabilities that they find.

Chapter 6: What's Your Cybersecurity Like Now?

The first step to improving your cybersecurity is to figure out what it's like now. You need to know where the strengths are, where the weaknesses are, and who might be looking to exploit them. You also need to accurately define what it is you have that needs protecting in order to understand what level of a target you are.

First thing's first, what is it that actually needs protecting?

It's important to understand what things you have that need to be protected and how appealing they are to attackers. If you're running an accounting firm or another company in financial services you need to understand that all of your files are at risk of attack. This means that you need to be much more vigilant with them and ensure that you use the latest anti-malware software, firewalls, up-to-date devices, and best cybersecurity policies.

If you're just worried about your personal cybersecurity, the same thing applies. If you're someone who doesn't use social media and keeps all of their money in cash under their mattress, you don't need to worry that much about cybersecurity. If you use online banking, lots of social media platforms, and regularly trade in cryptocurrency, you're a much more likely target for a potential attacker.

It's also about how you handle yourself when you use these devices. For example, if you regularly download pirated software and games, your risk is a lot higher. The same goes for using illegal streaming sites, clicking on phishing emails, and browsing the dark web. If you tend to ignore security updates for your devices, this is also going to increase your risk.

This also depends on who knows you and who might want to do you harm. If you're a celebrity or the CEO of an important company, there's a much higher chance that people are looking to infiltrate your devices. If you're simply a university student who goes to their classes and doesn't engage in any illegal activities, it's unlikely that you're as big of a target.

That said, cyber attackers are willing to go for anyone if they think it will make them money and many forms of malware infect any and all available devices. Just because you're not a big target doesn't mean you shouldn't take reasonable precautions

Next, you need to consider the people that you're surrounded by. If you're in a business, the worst employees are the ones that are setting the standard of your cybersecurity. It doesn't help if nine of them are perfect, if the tenth one clicks on every phishing email that they see, your company has weak cybersecurity. This is why it's important to have strict policies in place and make sure that all of your employees are educated on them.

If you're running a business that you think might be quite high risk, it's definitely a good idea to get a cybersecurity professional in to perform a proper evaluation. While this book can give you tips and tricks, it isn't a substitute for someone who has years of experience in the field. Especially not if they can take the time to spend hours poring over every inch of your cybersecurity, finding weaknesses in your systems and checking to see which of your employees are susceptible to phishing attempts. If you can afford it there's nothing better than having a professional check all of your systems to ensure that they're secure.

If you're a personal user and you're using shared devices with your family, your security is often only as good as the weakest link in the chain. It doesn't help if you do everything right, if your younger sibling decides to try and download an illegal version of Fortnite, it's likely that your computer will get compromised anyway.

It's also important that you're aware of what outside devices you're bringing into your network. Perhaps you regularly take flash drives from friends in order to give them movies or series. Or you often bring work home so you bring a flash drive from work with those files on. You might regularly take your work to various locations and use free public Wi-Fi. All of these external devices pose a cybersecurity risk and the more of them that you expose yourself to, the higher the risk that one of them will contain malware.

The next important step is to look at what devices you have. Your cellphone, TV, desktop computer, laptop, Wi-Fi router, each one of these devices is a potential security breach. The same goes for any other smart devices that you have in your household including smart bulbs, smart security cameras, smart scales, smart

speakers, smart vacuums, and smart cars. Are they recently released devices with up-to-date software? Or are they old and obsolete?

As devices age manufacturers stop releasing security updates for them. A computer using Windows 7 or an out-of-date smartphone is a much bigger security risk than one with the latest software. Ironically this means that the most tech savvy people often have the least secure homes as there are so many smart devices in them that can result in a potential security breach.

Continuing in the trend of software, your risk is dependent on how out of date all of your software is. Not just the security updates for your operating system, but also the updates for all of your mobile applications, computer programs, and applications on your smart TV or smart watch. Any one of these being outdated can pose a significant security risk.

You also need to evaluate whether or not you're using websites and software that prioritizes your security. If you're using an unsecure email provider or a messaging service that isn't end-to-end encrypted, your communication definitely isn't very secure. Open-source software is also preferable as none of the code is hidden and it's relatively easy to determine whether or not it's safe.

It's also vital that you're using strong passwords. Like we spoke about in the previous chapter, you need to be using a unique password for each account. The password needs to have a combination of numbers, letters, and special characters to be as strong as possible. In addition to this, it shouldn't use common words like the names of sports teams or the name of your favorite sport.

It also shouldn't be something that's personal about you and can be found out easily like your name, your birthdate, your children's names, or the name of your pet. If you're using weak passwords or the default ones that came with your devices, you can be sure that your cybersecurity isn't in a great place right now.

The next important thing to check is the safety and security of your data. If you're backing it up, if it's encrypted, if you're using a password for the device it's stored on, and if other people can access it easily. If you are backing it up it's also important to make sure that your chosen cloud storage provider is keeping the data safe. This means that they have proper security procedures, but also that they aren't behaving unethically and selling your data to the highest bidder for advertising purposes.

Another important thing to consider is your physical security. If you have a hard drive that isn't encrypted but is kept in a locked room with an armed guard at all times, it's still relatively safe. If you have a hard drive that isn't encrypted that you carry on a train or bus with you every day, it has a much higher chance of being stolen and the data being leaked.

Your internet is also an extremely important part of your overall security. It's safe to assume that almost all of your incoming and outgoing electronic information is travelling via your internet. This goes through your router, and to your devices. If you're using a firewall to act as a barrier and check all of the incoming packets, detect intruders, and deny access, then your devices are already going to be significantly more secure. This is because most cyber attacks will be stopped before they can even reach the devices. That means it doesn't matter if you

The last part that you need to consider is your specific cybersecurity software. Are you using an adblocker? Do you use blockers that stop you being monitored when you browse the internet? Do you have antivirus and anti-malware software installed that performs regular checks on your computer? Do you use a password manager to generate strong new passwords and keep track of them? Do you have a firewall configured for your network?

If you answered yes to all of these questions then you have nothing to worry about. If you answered no to one or more then these are areas that you should be looking at in order to improve your cybersecurity.

Chapter 7: The Biggest Risks to Your Cybersecurity

Before we take a look at how you can improve your cybersecurity on specific devices, it's important that you know what the biggest risks to your cybersecurity are. Once you know what they are you can figure out which ones are most applicable to you. That should help you to understand which things you need to improve on and what you should avoid in future.

The first major risk that we'll be taking a look at is public Wi-Fi. Public Wi-Fi can be dangerous for a number of reasons. One serious problem is that it often fails to use encryption. We'll cover more about encryption and why it's important later in the book but for the purposes of this chapter it's enough to know that it makes it harder for other people to easily see and understand your data.

With public Wi-Fi it can be quite easy for attackers to eavesdrop on you, intercepting your communication in order to see what you're sending. This can easily result in them getting access to important information like your bank logins, personal details, or financial data.

Public Wi-Fi is also a common target for man in the middle attacks that we spoke about earlier. Because they're not encrypted, it's a good place for attackers to try and intercept data and alter it. This is similar to simple eavesdropping and having your information stolen, but they could also be sending you malicious information in return like malware or links to phishing sites instead of the legitimate ones that you're trying to access.

Public Wi-Fi networks are also often used by attackers in order to entice unsuspecting targets into connecting. The targets either connect because the attackers have used the same name as a legitimate network or because they're hoping for some free internet. Many people also have their devices set to automatically connect to open Wi-Fi networks which means the users may not even notice that their device has connected at all. The owners of this network who are the

ones launching the attack can then harvest the targets information or even hijack their sessions.

For these reasons if you're using public Wi-Fi, you should always try to use a VPN that encrypts your traffic and prevents other people from eavesdropping on it or easily conducting a man in the middle attack. As you can tell it's a serious cybersecurity risk and it should be avoided if possible. The data savings of connecting to a free hotspot if you don't need to use it regularly often aren't worth the security risks.

Another common avenue of attack is outdated software. A lot of the time you will notice that your mobile phone or computer receive updates that don't seem to do anything. These are usually security updates, patching underlying weaknesses and holes in the software as they are discovered. If you fail to update your software or use devices with hardware so old that it no longer receives updates, you're at a much higher risk of having an attacker exploit a vulnerability.

A major risk to cybersecurity that people are often surprised to hear about is adverts. Many websites that host legitimate content fail to also make sure that the adverts being hosted on their sites are legitimate. This leads to a variety of harmful types of advertising. The first type is simply scam adverts. Someone will pay for an advert that looks legitimate to be hosted on a well-known site like Facebook or Amazon. When users click on this ad, they are redirected to a website that also looks legitimate for a business selling goods or services. Once they receive the money, they either never provide the promised goods or do something like sign the user up to a debit order instead of a payment.

Malware is also often injected into legitimate advertisements. Instead of redirecting the person who clicks on the ad to a legitimate website, they will instead be redirected to a phishing site. When they try to log in to the fake site using their details, the attacker will then have all of their information in order to steal their account on the legitimate site.

Other types of adverts redirect to sites that automatically attempt to start downloads. These downloads sometimes just contain an application that you don't want. Other times the download is malware that attempts to install onto your computer. This could simply be adware or it could be something worse like spyware or ransomware.

Quite often you don't even mean to click on the ad but they trick you by moving where the button to close an ad is or by putting something else in front of it. This is a practice known as clickjacking and it can often lead to users accidentally downloading malware or revealing confidential information. These days it's best to avoid ads entirely as the potential good deals you might get are outweighed by the cybersecurity risks.

Another major risk is social engineering. It is estimated that 98% of cyber attacks use social engineering and 82% of cyber security breaches are caused by human error. As computer security has become more advanced, attackers need very advanced skillsets in order to exploit vulnerabilities and develop malware that won't be detected. It's much easier to focus on the human element, which is usually the biggest risk in any cybersecurity network.

An example of an attack that purely exploits human error is leaving USB flash drives in locations where they will be found. The attackers simply buy a bunch of cheap USB flash drives and drop them in places like malls or outside a particular workplace whose security they want to breach. This is known as baiting.

When people see these lying around, they're either curious about what's on them or they decide that they want a new flash drive for themselves. When they plug it into either their personal or work computer, the drive autoruns malware. Many computers are set to not autorun files so the attackers will name the files something like "Confidential Employee Information" or "Fast and Furious 11 1080p". Once again, the human element is exploited as users opt to click on files that they know they shouldn't. These files then run the malware installation.

Another example of social engineering is attackers putting up a legitimate looking sign notifying people that the number for an IT helpdesk has changed. People then call this number hoping to get help with whatever technology issues that they have. The attacker posing as a helpdesk employee asks them for their login details and other personal information like their ID number.

The victim then willingly hands over their details, giving the attacker everything that they need in order to access their account.

Another major risk to cybersecurity is online privacy. Many people can't be bothered to maintain their privacy online. If a website asks for permission for cookies, they

automatically click “yes” just so that the popup disappears. When apps ask if they can collect their information, they don’t bother to untick the box that allows them to do so. They leave their social media accounts on public and allow their data to be collected for various forms of targeted advertising.

All of this makes it easier for attackers to conduct social engineering attacks. As people become more used to ignoring the fine print on installations and leaving boxes ticked without reading them, life becomes easier for cyber attackers.

As a joke a video game retailer in the UK once added a clause in their terms and conditions that said people who bought a game from them would surrender their immortal soul to the company. Thousands of people accepted these terms and conditions without reading them, proving that they don’t check before agreeing to online conditions.

If people are willing to agree to downloading your malware as part of another installation, it isn’t nearly as much work to get it onto their devices. Antiviruses can’t stop the users from installing any new programs at all, and if they willingly put it on their computer and consented to all of the attackers’ terms, technically the attackers didn’t even do anything wrong.

To summarize this chapter, people are the biggest risk to their own cybersecurity. Most breaches are caused by clicking the wrong links, falling victim to scam advertising, believing phishing emails, and installing programs without properly checking what they’re allowed to do and what permissions they require.

It’s much more likely that attackers will try to exploit you directly than a complex loophole in an outdated program’s code that requires extensive coding knowledge. The easiest and most effective attacks involve social engineering. This is why educating yourself on cybersecurity and understanding how different types of attacks happen is so important.

Chapter 8: Improving Cybersecurity on Your Phone

Almost everyone has a mobile phone. In fact, it's estimated that over 90% of the world's population are mobile phone users. And of those mobile phones, over 90% of these are smartphones. Smartphones are amazing devices; they combine the functionalities of a traditional mobile phone like calling and texting with the functionalities of a computer like browsing the internet and reading emails.

They can do almost anything, allowing users to browse the internet, take pictures, listen to music anywhere, watch videos, and communicate with people all over the world! They're truly a marvel of modern engineering.

They typically use touch screens and run either Android which is originally based on the Linux operating system for computers, or iOS which is based on macOS and stands for iPhone Operating System. The better devices are packed full of features including Bluetooth, Wi-Fi, biometric sensors, multiple different types of cameras, OLED screens, and wireless charging.

But with all of these features come downsides. They're much more complex to use than "dumb" phones and older people or those less technically proficient often struggle with them. They're expensive, they're easy to break, and if you use them regularly, they generally need to be charged every single day. This means that most people end up replacing them every 2-3 years, not just because they want a new model, but also because the old one is often already broken or damaged.

Because of this most manufacturers only keep providing smartphone updates for around 2 years and security updates for around 3 years. And by the time you buy your new phone it's often already a model that's been out for a few months or even a year. This could mean that your device is only actually receiving security updates for around two years of you using it. If you're someone who likes to look after your phone and use it for 4 or 5 years, you can see why this might be an issue.

We've come to accept some of these downsides and learned to live with them. After all, the features far outweigh these small issues. But there are other hidden

downsides that many users are completely unaware of. Smartphones don't usually run comprehensive security software because of their limitations in processing power due to their small size.

These devices are also packed full of your personal data. They often have pictures of you, your passwords, your banking details, and maybe even your fingerprints. Smartphones are also very commonly stolen items as they're small and high in value.

So, they're devices that are constantly connected to the internet, they often have confidential information stored on them, they're often connected to unsecured Wi-Fi networks, and they can receive communication from anyone who has your number. Their security software also isn't always up to date because manufacturers don't expect you to keep the phone for more than a couple years. In addition to this they can easily be lost, damaged, or stolen.

So how do you secure your mobile devices?

The first step is to make sure that you're choosing a mobile device where manufacturers place an emphasis on updates. Out of all of the major manufacturers Apple has been the best with updates, continuing to provide support for their mobile phones for around 5 years. This might be because they have very few budget devices and so their customers expect higher standards as most of their devices have higher price tags.

Other manufacturers like Samsung and Google have also committed to providing updates for their flagship phones for a long time. Google has provided 5 years of security updates for their Pixel phones and promises 7 years for future models. OnePlus, Xiaomi, Samsung also guarantee around 5 years of security updates for their flagship phones. The moral of the story is that if you want security updates for longer, and thus a more secure phone, spending more money on your device will get you just that.

But many people can't afford to buy flagship phones. The newest devices retail for large sums of money and often have similar features to much more budget devices. In this case it's probably best to buy a budget device and change it every few years. While this sounds like an expensive way to do things, it's often cheaper. If you sell your phone after two to three years while it still has value it can sometimes be a

better financial decision than keeping it until the screen cracks and it's almost worthless.

Not everybody is about to get a new mobile phone and you might not want to sell a perfectly good phone just because it doesn't get updates anymore. So, what else can you do to secure your device?

The second main thing that I would recommend is making sure that it has a password. This helps to keep all of your private data on your phone safe and secure. Now, you might be thinking, "I don't have anything to hide; why do I need a password?" If your phone does get stolen a password can be a very useful deterrent to stop thieves from misusing your personal data or pretending to be you and attempting to steal money from your loved ones.

A password will also make it more difficult for thieves to access your banking app on your phone and make unauthorized purchases or transactions. If a thief can access your phone settings, they can also easily disable settings that can make it easier for you to recover your stolen device or lock them out of your accounts.

It also prevents people from installing malware or tampering with your device while you're not looking. It's also important that you use a strong password or biometric access rather than something like "1234". We'll cover how to make strong passwords in a later chapter.

Another thing that's important is to make sure that the "Find My Phone" setting is enabled in your device's settings. If your phone is temporarily lost or even stolen it can help you or preferably the authorities to find the device and catch the criminals who took it. This feature also allows you to remotely erase the device, removing all of your personal logins and data from the stolen phone. This can be incredibly useful in keeping your data safe and saves you the trouble of having to change all of your passwords.

It's important that you only download applications from trusted app stores like the Apple App Store or the Google Play Store. While it may be tempting to source applications from other places, it's one of the easiest ways to accidentally end up with malware on your phone. If you are downloading apps from other sources, always make sure to double check user reviews and other info to make sure that it's as safe as possible.

Another crucial thing is to make sure that the apps you're using on your phone aren't compromising your privacy or safety. The messaging app you use should be end-to-end encrypted like Signal or WhatsApp. You should also check the permissions and privacy settings on your apps to make sure that your data isn't being used for anything that you wouldn't want it to be. If left unchecked many popular apps will collect as much of your data as possible and sell it to advertising companies.

While some guides or so-called experts will advise that you install antivirus software on your phone, it's largely a waste of time for mobile devices. These apps don't have access to the root files of the phone that they need to function correctly. This means that they're mostly for show and the companies are simply looking for a way to make money. At best the apps will do nothing, at worst they'll drain your battery and show you more adverts.

Adverts have also become a security risk as many companies aren't properly checking the ads that they host. Just because a well-known company like Facebook allows the ads to be put on their site doesn't mean that the ads are legitimate. Many ads hosted on otherwise legitimate sites or marketplaces are scams.

For this reason, I believe that it's better to avoid ads entirely. By using a free private DNS like dns.adguard.com, you can stop most ads from ever showing up on your device. This isn't an app or anything that you need to install, it's simply a setting that most phones have. You select the option to use a private DNS under settings and then put in dns.adguard.com or whatever other private DNS you want to use. This also helps to save you data as your phone is loading less ads and it's one of the most effective ways to stay safe on a mobile device.

Aside from these important steps, mobile security is largely about being smart with what you do. If you see a website, application, or advert that looks suspicious it's best to simply avoid it. Installing mobile apps from unknown sources is probably the single biggest risk to your phone. Mobile devices are naturally less susceptible to viruses than standard computers so if you follow this advice, it's highly unlikely that you'll have any issues with malware.

As for your accounts, data, and files we'll cover specific ways to keep these safe in later chapters.

Chapter 9: Improving Cybersecurity on Your Computer

One of the easiest ways to improve cybersecurity on your computer is to make sure that all of your software is updated. First and foremost, you should make sure that your actual operating system, whether it be Windows, Linux, or MacOS is updated and set to always update to the newest security patches.

Generally speaking, most computers that use either Linux or MacOS are more secure than their Windows counterparts. This is for multiple reasons and not necessarily because Windows has worse security as an operating system. The first reason is that the majority of computers use Windows. If an attacker is deciding what platform to develop malware for, they're probably going to choose the one with the most users. This increases the number of potential targets.

Both Linux and MacOS are also more focused on using an app store. While Windows does have an app store, most applications are obtained by simply finding them online and downloading them. This opens users up to accidentally downloading from the wrong sources.

It's important to understand that Windows has made significant security improvements over the years, with Windows Defender getting more secure and Windows prioritizing regular security updates. The operating system itself is secure enough that switching to an alternative is probably not worth the time or effort that it takes if your sole reason for doing so is security.

Like with mobile phones, adverts are also a big security risk for traditional computers. Ads on the internet don't usually go through much of a screening process and the companies hosting them don't care whether or not they're legitimate. Plenty cyber criminals buy advertising in order to try and convince users to buy illegitimate goods or download malware.

For this reason, having an adblocker helps not only with your overall browsing experience, but also to keep you safe. In fact, the FBI themselves recommend that users should be using ad blocking software.

The best ad blocker is widely accepted to be uBlock Origin. It's free, open-source, and available as an extension for most major browsers. It doesn't just stop you from seeing most ads, it actually stops the ads from running entirely. This reduces your data usage and actually allows your computer to run faster. This is because it's not using any resources to load or play the adverts, essentially your computer never "sees" the adverts at all.

It's also vital that your computer has a password in order to prevent unauthorized access. This should be a well-made and hard to crack password. In the next chapter we'll cover exactly how to make sure all of your passwords are secure.

The problem is, getting around passwords to access a user's files is actually remarkably easy if you have physical access to their device. All you need to do to access a Windows PC's files is to boot from a USB running another operating system like Linux. This completely ignores the Windows password permissions and allows you to edit and view the files. While this can be useful if you forget your password, it definitely isn't secure.

This means that if you want the files on your personal computer to be secure, you need to use an encryption tool. These tools can make sure that even if someone has your hard drive, they can see that there's information on it, but they can't actually see what the information is.

If you're worried about the people you know compromising your files, using your account logins, or looking at your information without permission and you have to share a device with them, creating multiple accounts on the device can be helpful. This means that when they log into their account they can't access any of your files, saved passwords, or logins. If they're someone who isn't very good with technology like an older relative or a younger sibling, you can also restrict their account from installing software. This helps to ensure that they don't accidentally install adware or malware onto your computer.

If all you want to do is keep your files from being snooped through, Windows also has a way to hide files. Certain system files are hidden by default so that normal users don't have to sort through them or get confused and accidentally delete the wrong thing. If you right click on a file or folder and open the properties tab there is

a checkbox that allows it to be hidden. Whilst this is by no means secure from cyber attackers, it's usually enough to keep friends or family from snooping.

Two-factor authentication can also be useful to prevent unauthorized attempts to login to your accounts. The first factor of the process is that you, the user, provide a password or PIN in order to login. This is generally known as the knowledge factor as you need to remember or know the information.

The next factor is that you use something else to finish the login. This is sometimes a security token that generates a code, but it can also be an application on your mobile device, or a smart card. This is known as the possession factor as it's something that you must directly have in your possession.

This means that if someone wants to gain access to your accounts via your computer, they need to have both your device and the second method of authentication. Whilst it's not impossible for some attackers to achieve this, it helps to hinder their efforts and they will often simply switch to an easier target.

There's also three-factor authentication, which uses what's known as the biometric factor. This means that in addition to the other two factors you need either your fingerprints, facial recognition, or a retina scan. This is even more secure than two-factor authentication but as you can imagine it isn't always the most practical solution. Three-factor authentication is generally only used when dealing with large quantities of money or very confidential information.

You should always make sure that your device's default antivirus and firewall are also enabled and configured. The antivirus helps to keep malware and other harmful software off of your device. It can sometimes also be beneficial to install third-party anti-malware software like Malwarebytes.

This software runs alongside your antivirus with no conflict but can be more effective at scanning and removing all traces of malware. This includes worms, trojans, rootkits, spyware, ransomware, and a whole lot more. In order to make sure you don't forget, it's usually best to schedule regular scans.

How often should these scans be?

That depends entirely on what kind of risk you're putting yourself at. If you regularly take external devices from people and frequent websites with illegal or unregistered



software, then you should be scanning daily. If you're someone who doesn't install a lot of new software on your computer and you're otherwise reasonably safe, then weekly or even monthly should be sufficient.

You should also be careful about the emails that you receive. It's best to avoid signing up to lots of unnecessary mailing lists and if you receive emails claiming to be from your bank or another sensitive organization, you should always make sure to check them carefully.

You can expect phishing emails to have spelling and grammar mistakes, use less professional language, and generally look a bit "wrong". They'll also either directly ask you for your details or try and redirect you to a site where you input your details.

Just like with mobile phones, the most important way to ensure your security is to be smart about what you do on your computer. Making sure not to download files from questionable sources and not to unlock your device for anybody is important. If you see software or people promising things that seem too good to be true, that's probably because they are. If you follow all of these policies that we spoke about, you can be sure that your computer is safe.

Chapter 10: Making and Using Better Passwords

Almost everyone knows what a password is. It's a secret word or phrase that you need to use to gain admission to a place. That place might be a fort that you made with your friends as a child or it might be your bank account. They've been around for literally thousands of years and were used by many ancient civilizations, including the Romans.

They were a way of making sure that the person they were speaking to was on their side and had the proper authorization to be entering a place like an army camp or watch tower. In this way, very little has changed.

Computers have been using passwords almost since their inception and in 1961 they were already being used at MIT (The Massachusetts Institute of Technology) on one of the first computer systems that allowed for multiple users. Usually when it comes to computers and online sites, each account has a specific password that is linked to it in order to login. This means that almost everyone has a password, or hopefully multiple passwords.

But what should it look like? Does it need letters, numbers, special characters, or all of them?

To start with, your passwords should always be unique. This means avoiding things like "123456", the literal word "password", "admin", "qwerty", and "111111". When a survey was done in 2016 it was discovered that the 25 most commonly known passwords made up over 10% of people's passwords.

Using a password like this makes you an incredibly easy target for attackers. They will usually run passwords through a password list. Even Wikipedia has a list of the 10 000 most commonly used passwords freely available for you to look at. It might be worth taking a look at this list to give yourself an idea of what not to do. A quick google search can even find you a password list with over 10 million passwords listed and freely available for ethical or unethical hackers to use.



If your password is anything very common like the ones mentioned above, it's guaranteed to be on that list. Our goal is to make gaining access to our sensitive information and accounts as difficult as possible. Using a password like the ones mentioned above is making it as easy as possible for attackers.

The next step is to make sure that your password isn't using common words and phrases. This includes things like your name, date of birth, children's or parents' names, names of your pets, and other personal details. If attackers decide to use social engineering it's highly likely that these will be some of the details that they find out about you first. If it's freely available to see on your Facebook or Instagram account, it definitely shouldn't be a part of your password.

The same goes for names of popular sports teams like "Yankees" or places like "Amsterdam" as well as popular characters in shows and movies. Many people also seem to like using their favorite sport as a password. You can be sure that you were not the first person to come up with "cricket123" and you definitely won't be the last.

Another way to make sure that your password is more secure is to make it longer. Longer passwords are generally more secure as they can't be cracked as easily through brute force attacks. For this reason, a password of around 12 or more characters is best. Common wisdom among cybersecurity professionals is that a long password can beat a complex one, simply because they're harder to brute force.

It's also a good idea to avoid using repeated character or sequences. Computers are a lot better at sequences and patterns than people and if you're using sequential numbers like "123" in your password to try and make it longer, it's not helping to make it much more secure at all.

It's also good to vary your password by using uppercase letters, lowercase letters, numbers, and special characters. The larger pool of different character that you're pulling from, the more difficult it is to brute force your password. Special characters and numbers also aren't in the dictionary which makes standard word attacks more effective.

That said, you shouldn't use obvious substitutions when trying to add special characters or numbers. Using a dollar sign for "S" or substituting an "E" for a "3" isn't very effective. It's better to simply use random symbols and numbers mixed in between letters.

The same goes for what's known as password "walking" which is where people use characters that are in a similar order to the keyboard. For example, "q1w2e3r4t5" sounds like it's a complex password because it has numbers and letters all mixed together. But it's actually quite easy because they're in order and it's actually in the top 200 most common passwords. This means if you're using it, an attacker can crack your password within a matter of minutes.

Using unusual words or acronyms is also an easy way to help you make a password if you're creating one yourself. If you speak a language other than English those words are less likely to be on standard password lists as most passwords are usually in English. Combining an unusual word or words in a language that isn't English with some randomized special character and numbers mixed in can create a very strong password.

It's also very important that you use a unique password for each device or account that you log into. That way if there is a security breach on the company's side and one of your passwords has leaked, hackers can't use that password to log into any of the other accounts associated with your email.

Another thing to note is that when there's been a security breach at a specific site that you use, you should change that password. You don't have to change all of them because you should already be using a unique password for each site or application. Remember not to change your password directly on the email or message that tells you there has been a security breach, because this could easily be a phishing attempt.

Instead, you should independently visit the official site and/or their social media and check to see if there is an official notice of the breach. If there is, make the changes directly on the organization's website. Doing it this way makes sure that you don't accidentally give an attacker your details.

Having this many passwords that are unique and use all kinds of numbers, symbols, and letters can be difficult to keep track of. If you use lots of different online shopping services, social media apps, and other accounts, you can easily end up with more unique passwords than you can remember.

This is where a password manager comes in. Password managers can generate as well as store complex passwords for as many accounts as you need. This makes it so that you only have to remember the passwords for your device and the password



manager that you use. You can use the same password manager app on your mobile phone as on your computer, making it easier to keep track of all of them and log into sites.

Password managers never use very common passwords that can be found on lists as they randomly generate them using a combination of numbers, letters, and special characters. If you're following these practices, it's almost impossible for attackers to guess your password using dictionary attacks or brute force it.

This means that they have to resort to other tactics like phishing in order to try and convince you to put your secure password into the wrong sites. Another benefit of password managers is that they won't attempt to automatically input your details into a phishing site. The password managers immediately pick up that the site is different. So, if you can't figure out why your password manager isn't automatically signing you in to Facebook, it might be because you're actually trying to sign into a phishing site.

In summary you need to make sure that you: One, use unique passwords for every site. Two, don't use common words to make your passwords. Three, make sure to use a combination of letters, numbers, and special characters for maximum strength. Four, don't use sequences of numbers like "123" in your password. Five, don't use obvious substitutions when making your passwords like a dollar sign for the letter "S".

And if all of that's too much to remember, just sign up for a password manager and let them do it for you! If all that you remember from this whole book is these tips on how to improve your passwords, it will have been time well spent.

Chapter 11: Why Does Your Data Need to Be Kept Safe?

A large part of cybersecurity is making sure that your data is kept safe and secure. Now, I realize that some of you might be thinking, “I don’t have anything to hide, why does it matter so much.” It’s important to realize just how much information most people have stored on their electronic devices and accounts. Medical records, unique identifiers like your full name, address, and identity number. Your financial records like your pay slips, loans, bank account number, and bank statements.

As well as personal photographs and videos of you and other friends and family. It’s your responsibility to keep your data safe, but also that of your parents, children, significant other, and the company that you work for. There are a number of reasons why most people should probably be stricter about this data’s security.

The first one that’s very common but sometimes overlooked is identity theft. It’s easy to forget just how easy it is to impersonate someone over the phone using a few of their details. In addition to this there can be lots of identifying information on a personal computer. Many people have a scanned copy of their ID, accounts for things like water and electricity, and other documents that make identity theft incredibly easy.

The criminals involved can then use this information to open fake bank accounts in your name, apply for credit, or make changes to any of your existing accounts like internet, water, and electricity.

This leads us to the second reason why you should protect your data, it can lead to significant financial loss if you don’t. Attackers have no regard for your personal finance and they will quickly run up as much in expenses as they possibly can. A significant security breach can lead to massive financial loss if not caught quickly enough.

The financial loss can also be long lasting. Your credit score could be negatively affected if an attacker uses your details to open new accounts in your name and never pays these accounts. And, you can be sure that they have no intention of ever

paying any of these accounts. If they intended to pay them, they wouldn't have a problem opening them in their own name. In addition to this the costs of recovering from a data breach are usually significantly more costly than simply preventing one in the first place.

Another reason is that attackers will quickly and easily damage your reputation. They might leak private pictures or videos of you if you don't agree to pay them a certain amount of money. Even if you do pay them, there's no guarantee that they'll keep their word. They can also message your friends and family members using your email account or cellphone number. This can lead to their security being breached.

They also might have trusted the message because it came from you and they trust you. This can then spread to their other contacts. Ultimately, even if it isn't really your fault, if the security breach came from your number, they'll hold you responsible for it.

Another reason to protect your data is that you might end up losing it if you don't. If you find yourself the victim of ransomware or a deletion attack, you could end up losing years of pictures, journal notes, your favorite movies, and maybe even important work.

It's also less effort to keep your data safe than it is to deal with a security breach. Once you've had a major breach, you need to change everything. You might need to get a whole new phone number or email account if you've been locked out of yours. You'll also need to update all of your passwords, especially to sensitive things like online marketplaces and your banking login.

This might not be enough as the authorities could have picked up the illegal activities happening in your name. If they're serious, like accounts being unpaid, you will need to open a case with the police and spend time to ensure that they have all of the relevant details.

It's not always easy to prove a breach of cybersecurity and for authorities and credit institutions it's much easier to simply hold you responsible. If they acknowledge that it was caused by an attacker, they might never get the money back. If they refuse to acknowledge it was an attacker and insist that you pay as it's in your name, they have a much better chance of recovering their money.

There's always the chance that data other than your personal data is stolen. You might be signed into a work email account or sometimes use your personal computer for dealing with work information. Or perhaps it's a work computer that has had information stolen. If you manage to accidentally be responsible for a leak of customer information, they will lose trust in you and the business itself. It's common after data breaches for customers to switch to a different company as they do not feel that their personal information is being adequately protected.

Their data that has been stolen from you can often impact their security when using other accounts or in their daily lives. Most people don't follow the optimal cybersecurity procedures when it comes to using unique passwords. In fact, studies have found that only 12% of people make sure to always use unique passwords for all of their accounts.

This means that if there's a data leak, there's a very high chance that the customer whose data has been leaked has used this password for at least one other account. This means that even if you have a data leak for something relatively insignificant like a news site, you could still be indirectly responsible for customers losing money as a result of the leak.

Even if your data isn't lost as a result of a security breach, and it's simply caused by a fault hard drive or other technology issue, it still sucks to have to deal with. If it's not properly backed up you could lose significant amounts of work. This results in you having to spend time redoing it, and as I'm sure you know, time is money.

Ultimately all of these consequences can result in unnecessary emotional distress and anxiety. Knowing that your personal information or other people's personal information that you were supposed to be looking after could be leaked as a result of your negligence is a terrible feeling. As you can tell a serious breach of your data's security is a massive issue and unnecessarily costly in a variety of ways. Because of this, keeping your data safe and secure is an investment.

Chapter 12: Keeping Your Data Safe and Secure

After the previous chapter you should have a good understanding of why it's important to keep your data secure. This means that you're ready to make some necessary changes in order to ensure that it's something that happens. But what changes? How should you make sure that your data isn't leaked or lost?

The first step is to make sure that all of your important data is backed up. This is a good way of ensuring that if your device is lost, stolen, or irreparably broken, you can still obtain all of your important information. There are a number of reasons why it could be lost including natural disasters, hardware failure, accidental physical damage, theft, and malware.

First, where are some places that you can back up your data?

For the purpose of this book, we'll be grouping together services that are both considered sync services and backup services. Syncing means that a file can be available in multiple locations, and changes will be reflected across those locations. Backups are only copied from one place which means if you accidentally delete a file, you won't lose it. This used to matter more than it does now, as now most sync services have a "trash" which is basically the online version of a recycle bin.

If you accidentally delete something it isn't too much trouble to get it back from either the recycle bin on your physical drive or the trash bin on your online syncing service. Some popular free storage services include Google Drive, Microsoft OneDrive, and Apple iCloud. It's worth noting that these services only give you the first few gigabytes free, so if you need more storage, you're going to need to pay for it. Storing data costs these companies and unless you're paying those costs, they're either selling your data or making a loss off of storing it for you.

It's also worth considering a physical backup. It can be more effort than an online one but depending on how often your data updates, it could be worth doing. A large storage drive for family photos and videos isn't very expensive compared to monthly

cloud storage. The important thing to remember is that you need to keep this storage at a physically separate location to where the rest of your data is stored.

Otherwise, if there's a natural disaster or fire, you could easily end up losing everything.

But how do you decide which data to back up?

You should back up anything that you consider important. To most people one of the most important things is often their family pictures and videos. There is no price that can be put on a video of your child taking its first steps or your last day spent with your parents. These files are often quite large, and it will take a while to upload all of them at first. However, once it's done, adding any new ones will be fairly quick.

Something that people often neglect to backup is a list of software licenses and product keys. It's easy to forget that a lot of the software on your device isn't free. If you end up needing to do a fresh install of Windows, Microsoft Office, or another paid program, losing your license keys and activation codes can be a costly lesson. It can also be useful to back up other software files containing your computer's configurations, especially if you have made customized changes. If you're someone who plays single player games offline, it can be nice to keep a copy of your settings and saved games in case anything like that is lost.

It's also important that all of your other passwords and login details are backed up. If you're not using a password manager that syncs automatically, make sure that the details are at least saved somewhere. If you simply leave your passwords on a sticky note on your desktop, aside from the fact that this isn't very secure, if they go missing it can be a massive headache.

The same goes for any other active projects that you're working on. If you're researching how to build a table or compiling a project for university, it's a massive amount of time and effort if those files are lost. Whatever it is that you're currently working on or might need to go back to in future, it's best to simply have it backed up.

To businesses, the things that are considered important are a bit different. Most businesses need essential documents backed up like contracts and other important employee documents, the registration documents for the company, and current



projects that are being worked on. It's also common to backup financial information like tax records, bank statements, and the actual file for whatever accounting software is used.

If any accounting information is lost it can be both painful and time consuming to get it back. For this reason, a lot of paid accounting software actually has its own backup built in. It's still better to be sure and back it up yourself as well.

Companies also need to back up other things like databases, websites, supplier information, and customer information. Imagine that you run a business selling hardware products. If you were to lose all of the information on your customers you wouldn't know which ones need to be contacted about sales, where the goods need to be delivered, or perhaps even how to contact them at all. For bigger businesses like Facebook and TikTok, the information they store on their users is worth a large amount. To lose this information would be absolutely catastrophic for their business model.

In short, if it's important, and it's on your computer, back it up! It's much better to spend a little bit on paying for extra cloud storage or a program that backs your data up automatically than it is to have to pay for all of your licensed software again. Or, even worse, lose important pictures and videos of your family members that you can never get back.

But how do you make sure that your data is also secure? You don't just want it to be safe from being lost. You also want it to be safe from being stolen, or snooped through.

The best way to make sure that your data is safe from the wrong people looking at it is to encrypt it. Now, you might be wondering, what exactly is encryption?

Encryption has been around for a long time and we have records of the ancient Egyptians using it as early as 1900 BC. This early type of encryption was used throughout Ancient Greece and Rome by the military. You may remember from one of the previous chapters that the Romans also used passwords as part of their military system. Encryption continued to be used throughout history and as we move towards more modern times, the infamous Enigma machine was used by the Axis powers (Germany, Italy and Japan) in order to encrypt their top-secret military communication.

Fast forward to today and encryption is now synonymous with computers and technology. Most communication over the internet is now encrypted and it continues to evolve in order to prevent eavesdropping attacks. You may remember from earlier that we spoke about WhatsApp, Signal, and Telegram being forms of encrypted messaging.

But, even aside from communication, encryption can be used to protect the actual data that is stored on your physical computer. As we already established earlier that passwords can be easy to get around, the logical solution is to encrypt all of the data stored on a hard drive.

That way, regardless of what operating system an attacker uses, they need the encryption key in order to view the actual data on your drive. This is not a flawless system but it adds an extra layer of security to your data.

Most modern operating systems like Linux, Windows, and MacOS offer built in encryption software. You can download an additional program if you want to, but ultimately the built in software offered is quite effective. There's usually a wizard that comes with this software that helps you with the process. You can choose which files or parts of your drive specifically need to be encrypted based on their importance. Depending on the tool you're using, you might also be able to choose the key strength and specific algorithm used.

This is a strong deterrent against most thieves who might steal your devices as well as your average snooping person or cyber attacker. It is not, however, foolproof as encryption keys have started to become vulnerable to brute force attacks. As computing power has increased, so has the strength of brute force attacks.

This has made some forms of encryption and keys less effective. It's important to always use a longer encryption key as they're harder to detect. This will become increasingly important as stronger and stronger computers become better at cracking encryption algorithms.

The good news is that very few people have access to these machines and if you're the average person, chances are they're not interested in you.

If you have any important data at all, you should absolutely be backing it up, either physically at a separate location or in the cloud. Encryption is less important for the



average person's data, but a helpful deterrent if you're dealing with data that's more sensitive or you think you might be a high-risk target.

Chapter 13: Making Sure Your Accounts Are Secure

If you've been paying attention to the other chapters, you might already have a reasonably good idea of how to keep your accounts secure. There's definitely some overlap between password management, data security, device security, and general cybersecurity practices. That said, it's important that we look at how to protect accounts specifically and also refresh some of the other practices so that you don't forget them.

The first step to having more secure accounts is using stronger passwords. Remember, these passwords should be unique to each account that you have. They should also use a combination of letters, numbers, and special characters for maximum strength. If you're not sure how strong your password is, rather just make it longer. The minimum password length is just that, a minimum. Ideally all of your passwords should be at least 12 digits long. If all of this sounds like too much to remember, simply use a password manager. It will create strong passwords for you and it saves you the trouble of having to remember all of them.

It also helps to enable two factor authentication. This means that instead of just being able to use your password, attackers also need access to your authentication device. If this is a physical device like a security token then they need to gain physical access to it in order to access your account. Alternatively, if it's something else like an SMS, they still need to gain access to your mobile phone or number in order to receive this SMS. Two factor authentication is significantly stronger than just a password, especially if you're someone who makes a habit of using weak passwords or reusing them.

It's very important that your email account remains secure in order to keep all of your other accounts secure. Out of all of your accounts this one should be the hardest for attackers to get into. Even if you don't have two factor authentication enabled on your other accounts, consider enabling it on just your email account. This is because your email account can be used to gain access to almost any other account that you have.

If an attacker gains access to your emails they can very quickly work out which accounts you're using based on which companies are sending you emails. They can then attempt to login to those accounts using your email password. If that doesn't work, they can simply request password resets on all of your other accounts. The password reset links will be sent to your email and they can easily change your passwords. This won't just give them access to your accounts, it'll also lock you out of them, preventing you from easily changing your passwords to regain control.

In order to prevent people from gaining unauthorized access to your accounts, your devices also need to be secure. It doesn't help to use an amazing password manager with secure passwords if anybody can walk past your laptop and open it. If you want your account password to be secure, the devices that they're stored on need to be secure. This applies to having passwords on your devices but also to ensuring their physical security. It's important that you don't leave your laptop out where it can easily be stolen. Unless you've encrypted the information on it, attackers can reasonably easily gain access to all of your files along with your saved passwords. These saved passwords will then give them access to your accounts.

You should also regularly monitor your accounts for any suspicious activity or anything else unusual. If the company who the account is with has a security breach it's important that you know so that you can change your password. In order to monitor this and any other suspicious activity, it's best if you have notifications for your accounts enabled.

You should also check what permissions your apps and accounts require. This isn't just about keeping the accounts secure, it's also about protecting your own privacy. For example, your calculator app doesn't need access to your location and your game apps don't need to be able to access your phone calls. Wherever possible restrict the permissions that these apps have. You should also check the permissions your accounts have for data collection in the account settings. Often the difference between an account respecting your privacy and collecting and selling your data is a simple checkbox.

The last, but definitely not the least, thing that you need to be aware of is social engineering. If your devices are secure and you have reasonably strong passwords then there is one other major avenue of attack. The attackers will try and gain access to your accounts by tricking you. You should always be aware of this and be careful

when receiving any communication requesting that you enter or change your passwords or anything similar.

For example, if you work for a large company and you're contacted by an IT person from your company, you should verify that their email address is a company one and not something like a Gmail account with your company name in it. You should also refrain from giving anyone your password over the phone or a text message.

Legitimate support staff will never ask you for your password as they can gain access to your account information from their own side without needing it.

When entering your login details for any account, always make sure that you're on the correct and legitimate version of their website. Also remember to be wary of online quizzes which can attempt to extract important information from you that can later be used to login to your accounts. If there's a quiz that it's any way related to a password or security question that you might have, it's better to avoid it entirely. The most important thing is always to be mindful of what you're doing online and who you're giving your credentials to.

If you follow all of the recommended best practices for passwords, two factor authentication, and device security, you're already significantly ahead of most people when it comes to protecting your accounts. If you also keep an eye out for phishing emails and other social engineering attempts, it's highly unlikely that attackers will manage to gain access to any of your accounts.

Chapter 14: Physical Security in a Cybersecurity Context

It's easy to think that there's no point talking about physical security in a book about cybersecurity. You might even be tempted to skip this chapter entirely, thinking that it's a waste of time. However, that couldn't be further from the truth.

As you've learnt from the previous chapter, there is no truly perfect way to protect your data if an attacker has access to your physical drive. Passwords can often be bypassed, encryption is becoming less effective as computers grow in processing power, and given enough time almost anything can be cracked. Don't get me wrong, all of these steps are helpful and I'm not saying that you should ignore any of them. But without physical security first, cybersecurity can be a losing battle.

Imagine that you have two laptops. They're both running Windows and are weakly protected, with only a standard password being used. From this book we've already learnt that all an attacker with physical access to your laptop needs to bypass this Windows password is a bootable USB drive running Linux. Now imagine that the one laptop is kept in a secure bank vault that's only ever opened by authorized staff. And the other you take with you to work every day on the bus, train, or walking. Which laptop is more secure?

Obviously, it's the one locked in a vault that attackers can't gain physical access to. Before the dawn of the internet, security for computers largely involved restricting physical access to them. Nowadays computers connected to the internet can be accessed from anywhere in the world, making it much easier for attackers to gain access to them without needing to physically be there.

But, despite all of our advances in technology, the easiest way for an attacker to get the data off of your computer is by having it in their possession. This means that one of the best ways to ensure your cybersecurity is still to ensure your physical security.

There are essentially two categories of devices. There are mobile devices, like your cellphone, laptop, smart watch, or tablet that you carry to different places. And then

there are stationary devices like your desktop computer, smart TV, and gaming console.

Stationary devices can still be stolen, fall victim to a physical data attack or be broken by a malicious intruder, but they're substantially easier to protect. If you have a lock on your door, an attacker has to get through that lock to get to your desktop computer.

The same goes for a fence, alarm system, and whatever other physical security you have in your home or office. Most of the time the only people who can reasonably easily gain access to your stationary devices are those who you voluntarily let into your home or office.

Mobile devices are inherently more at risk of a physical attack than their stationary counterparts. You might think that your phone or laptop is safer because you always keep it with you. But you need to consider how many places you take it to and how many different people you come into contact with. Every time you go shopping, every time you take a trip on the bus, every time you stop at a red light while driving, there could be someone waiting to steal your mobile device. This isn't to scare you, rather it's to make you realize just how unsecure the physical security of these devices truly is.

If you have secure data stored at a specific location there are easy steps that you can take to improve its physical security. You can add an alarm system, you can add extra locks on the doors, you can make sure only a select few people have keys to enter. If it doesn't need to be connected to a network you could even put the storage device into a safe. Perhaps if you're rich and you have too much money on your hands you could even build a concrete bunker and hide the device underground with someone guarding it 24/7.

None of this applies to mobile devices. Short of avoiding more dangerous places in town and hiring a bodyguard, there's a lot less obvious ways that you can make sure that your mobile devices are safe.

Now that you know what the risks to the different device types are, and why physical security is so important, it's time to take a look at some of the ways that you can actually improve it. I'm sure you've started to notice a pattern in this book; a lot of



cybersecurity is about how you handle yourself and the choices that you make. Physical security is no different.

We'll start with mobile devices as these are the ones usually at higher risk. The first thing that you can do to ensure that they're safe, is making sure that they never leave your sight when you're not in a safe place.

Are you going to the bathroom at a restaurant or coffee shop? Don't leave your phone on the table, put it in your pocket and take it with you. If it's something bigger like a laptop and you can't carry it with you, try and make sure that you're leaving it in as safe of a place as possible. If you leave it in your car, make sure that the car is locked and the laptop is hidden somewhere out of sight. While this isn't a perfectly safe solution, all of these little things help.

If your laptop has a small slot with a symbol of a physical lock next to it, you may have wondered what it's for. It's for a type of cable lock known as a Kensington lock. One end of the cable fits into the laptop and the other end can be wrapped around a desk leg or something else that's fixed. While these are far from indestructible, they can deter more casual thieves or slow down more determined ones.

Another part of physical security for mobile devices is making sure that they're safe from you. I know what you're thinking, why would I damage my own devices? The simple answer is, because it's so easy to do it accidentally. One accidental drop into a swimming pool or a completely smashed cellphone could easily result in data loss. This means that no matter how tempting it is to show off your shiny new phone with its clear glass back, it's better to put it in a cover instead. Mobile phones should also always have screen protectors to help prevent the screen getting scratched or breaking when you accidentally drop them.

As for laptops, they should always be kept in a dedicated protective laptop bag. This should fit the laptop snugly and it shouldn't have room to jostle around while you're moving. If you prefer to put your laptop into a work bag or backpack, consider getting a laptop sleeve for it. It should be as snug fitting as possible and will protect the laptop from getting scratched or damaged by anything else inside your bag. Laptops have gotten smaller with time so make sure that you check the exact dimensions of your device and the bag. Just because it says it fits a laptop with a

fourteen-inch screen and you have a laptop with a fourteen-inch screen doesn't mean it's going to be the right size.

Stationary devices are quite a bit easier to keep safe as by improving the safety of wherever they're located, you improve their security. If you live inside a complex with a security guard, your desktop computer also has a security guard looking after it. By installing proper security at your home or office you're also installing proper security for your smart TV or desktop computer.

As for stopping the devices from being physically damaged, it's important that you put them somewhere where they can't easily be knocked over and away from any water sources. Proper cable management can also help to keep them safe as people are much less likely to trip over a power cable and accidentally pull a TV or computer monitor onto the floor. It's also worth investing in an uninterrupted power supply, commonly referred to as a UPS, in order to protect your sensitive electronic devices from power surges.

Physical security will never be perfect. There will always be a safer country that you could be living in to protect against theft. There's always the risk of natural disasters like flooding, or a fire destroying your electronic devices.

This doesn't mean that you shouldn't try to improve your physical security, just that you should understand the risks and do your best to mitigate them where possible. It's also why things like backing up and encryption that we spoke about earlier in the book are so important. If you're doing your best to improve your physical security, and backing up all of your data, you can be sure that you're in a reasonably secure position.

Chapter 15: Improving Your Business Cybersecurity

We've already touched on various aspects of business cybersecurity in this book. Knowing different types of attacks, knowing who the attackers are, and using strong passwords are all things that can help you to improve the cybersecurity of your business. But about business specific cybersecurity practices? Don't worry, we'll cover those now.

They're important to know because the risks for businesses can often be higher, and the consequences worse, than those of individuals.

A lot of business cybersecurity involves assessments, plans, and procedures. When it comes to your private cybersecurity you know that things are being done right because you're the one doing them. When it's a business, there needs to be a policy for everything. Employees need to know what they can and can't do with work information, what devices they're allowed to use for work, and what information is most sensitive.

Most modern cyber-attacks rely not on finding weaknesses in the actual hardware or software, but instead on finding weaknesses the actual people. This means that in a business a large part of your cybersecurity resources and time won't be spent on patching complex vulnerabilities and fighting back against hackers. Instead, it'll be spent educating employees and making sure that they're not unintentionally compromising your systems.

Just like general employees need to know what to do when it comes to cybersecurity, the person in charge of it at the company needs to know what to do if something goes wrong. There must be a comprehensive incident response plan to be used if something goes wrong. This will include things like how to identify the threat, how it will be contained, and how to recover from the incident. If you're looking to set up one of these plans, the chapters on identifying and recovering from cybersecurity breaches will be highly beneficial.

You should also conduct regular audits. These should identify vulnerabilities, assess risk, and ensure that the employees are actually implementing the security measures. This is especially important when technology changes or new employees are brought in who aren't yet familiar with the company's policies.

Employees should also have what's known as role-based access control. This means that each employee is restricted only to the resources and information that they actually need to perform their role. If you have a factory, your production manager doesn't need access to all of your financial reports and accounting files. If they need access to any financial at all, it's only the ones actually pertaining to the production department.

This goes hand in hand with identity and access management. There needs to be strong authentication protocols in place to ensure that only the individuals with the correct authorization are accessing sensitive data. Regular reviews should be done to ensure that if an employee changes departments or positions, they have access to what they need for their new position.

They should also not have access to files and information from a previous position that they're no longer working in.

Businesses focus a lot on supply chains, and cybersecurity in businesses is no different. It's important to ensure that both the suppliers and vendors you deal with aren't placing you at significant risk of a cybersecurity breach. This means that you should check that their cybersecurity policies and standards are up to scratch. In addition to this, you should always use secure communication when dealing with suppliers and vendors. End-to-end encryption can help to make sure that no sensitive information is intercepted or leaked.

Sharing is also an important part of the business cybersecurity process. If you set up a system where you share cybersecurity information with peers in the industry it can help you to detect threats before they're an issue for your company. You can also discuss best practices with them and figure out what everyone else is doing in the industry in order to determine what works and what doesn't. This is a form of continuous monitoring and it helps to properly identify and respond to emerging threats.



If you're worried about your business's cybersecurity or you don't feel up to the task of some of these solutions, it's probably a good idea to hire a professional. They can help with things like penetration testing, finding where vulnerabilities are in systems, developing incident response plans, and much more.

For a business cybersecurity isn't just an expense, it's an investment. Like regular maintenance of machinery and buildings, your cybersecurity also needs regular maintenance. Employees need to be trained regularly, new equipment needs to be made secure, and files need to be backed up. Security isn't a product, it's a process that needs to be looked at regularly.

Physical security is another vital area of business cybersecurity. Businesses are much more likely to have a number of unknown parties coming and going than your home. Every customer that comes to a business premises is a potential security risk. It's important that they can't easily gain access to any of the computers or other infrastructure.

This means that all computers should be password protected, but also that routers and access points shouldn't be easily accessible from areas where customers normally are. Using access control, security cameras, and securing all devices with access to sensitive information is an important way to ensure that nothing is compromised.

We already covered backing up data as a section but it's often even more important for businesses than it is for individuals. The chances of a malicious person deleting files are a lot higher when you have to sometimes fire staff. Small and medium sized businesses are also very vulnerable to ransomware and are often the most common targets.

Being able to simply restore files from a backup is a very important thing in a business. After all, you won't just be losing pictures of your dinner and pets. You could be losing years of financial information, employee information, and vital company files.

This is also why it's a good idea to make sure that your internet is secure. Even with cybersecurity training, there's always the chance that a new employee or one who isn't paying attention clicks on a malicious email link. Email scans are a useful way of ensuring that emails aren't being exposed to phishing attempts or being used for

personal information which would also increase their risk. A DNS implemented at a network level is a good way of making sure that a large portion of ads and malware is being filtered out before it even reaches the users. You can also block specific websites and types of websites with a firewall to improve productivity and keep your network safer.

Things like piracy sites, social media, and maybe even video streaming can help to ensure that your network is safer and runs faster for the things you actually need it to be doing. After this you can also add software to each device that blocks ads, restricts cookies, and generally reduces the amount of tracking and malware that an employee could be exposed to.

Generally speaking, the less access and permissions your employees have, the less damage they can do. If they don't need to install software for their job, they probably shouldn't be able to. If they're not managing your company's social media, they probably shouldn't have access to it on their work device. While some staff won't enjoy these policies, they're important for keeping a network safe and running optimally. After all, if they want to be browsing Facebook or TikTok during their breaks, they're perfectly capable of doing it using their own data and mobile phone or personal laptop.

In business cybersecurity it's important to make sure that you're complying with regulations. We won't cover the specific regulations as that depends on your region, but almost everywhere has some cybersecurity regulations. These are especially important when it comes to handling customer and employee data. If you mismanage your personal business data, you could potentially have it leaked. If you mismanage your customer or employee data, you could easily end up facing a costly lawsuit. When it comes to this kind of data you need to make sure it's even more secure than you think it needs to be.

If you believe that your business is high risk, you could also look at something like cybersecurity insurance, also known as cyber liability insurance. It's most important for businesses that store or produce sensitive information.

For example, if you're running an online store and customers are giving you their street addresses, phone numbers, and names. This type of insurance covers losses that might be incurred as a result of data destruction, hacking, data extortion, and



data theft. Depending on the policy you get it might also cover your legal costs if you face a lawsuit as a result of a cybersecurity breach. Like with any other type of insurance whether or not you need it is largely dependent on your risk profile and if you can afford to recover from the worst-case scenario.

Chapter 16: Other Cybersecurity Tips and Tricks

We've already spoken about a variety of ways to improve your cybersecurity, but in this chapter, we'll cover some specific tips and tricks. Some of these didn't fit into any of the other sections but they were too useful to leave out. A lot of them are specific and easy changes that you can make in order to improve your cybersecurity affordably and effectively. Even if you just start with one or two, they're a great way to improve your cybersecurity.

The first tip seems simple, it's to be careful with your personal information. There's a common saying that comes from television in the 1970s, "If they're not charging you for the product, then you are the product." This is especially true when it comes to most modern internet services. Think of all of the different great things that you can use online absolutely free of charge.

There are social media platforms like Facebook, Instagram, and TikTok. None of these sites charge you anything to use them. There are search engines like Google which are also free. The same goes for video streaming with sites like YouTube providing completely free content for you to watch. Even GPS is now free, services like Google Maps and Waze don't charge a thing.

Every site that you open talks about enabling cookies. Cookies are essentially small files that remember what you do online. Some of them remember things that help you like what font size you like or what language you want the site to be in. They could also remember what's in your cart so that as you navigate from one part of the site to another, it remembers all of the things that you've added.

But other cookies are a little less helpful to you. They store information about you as you access different websites so that they can figure out how to best advertise to you. They track you as you browse the internet, collecting as much personal information as they possibly can.

What does all of this mean? It means that you're the product. Ultimately these are all companies with the goal of making money. If they're not receiving money from the

people using the services, they must be getting it from somewhere else. If they weren't, they would all be terrible businesses. This means that every piece of information you give them is going to be used in some way.

If Google sees that you recently used your GPS to go to a hardware store and that you've been using YouTube to watch videos on how to put up a fence, they have a pretty good idea of what kind of project you're undertaking. You can expect them to use this to advertise to you. It's up to you to decide how many of these services you want to use and which ones you feel are worth the loss of privacy. Ultimately the most important thing is that you're aware that it's happening and mindful of which companies you're giving your data to.

Likewise, it's generally a good idea to avoid oversharing of personal details. While it might be tempting to have everything about you on your social media, it's better to avoid this. The less people can easily find out about your date of birth, home address, and phone number, the better.

This tip is a small one, but always remember to change the default passwords that your devices come with. Many Wi-Fi routers for example have the default login password and username both as admin. Also, if you leave your Wi-Fi password as the default one, all anybody has to do to get your password is look at the bottom of the router. That's in no way a safe and secure network. Plenty of other devices have the default passwords easily available online. If your device is still using the default name and password, it's often a simple google search away from being accessed by someone without permission.

Another easy way to improve your cybersecurity is to use a Virtual Private Network, usually just called a VPN. A VPN protects you in a number of ways, all while being relatively inexpensive. There are even some great free VPN providers but they usually have limited settings, speeds, or only allow users a limited amount of data. The first important thing that a VPN does is that it encrypts your internet traffic. Everything travelling between your device and the VPN server is completely encrypted.

So, if someone intercepts your traffic, they can see that you're connected to a VPN but they can't see what you're doing. This stops your ISP from spying on you and

also helps to keep you safe when connected to public Wi-Fi or another person's internet.

The second way that a VPN helps is that it hides your IP address. An IP address is a unique identifier that can allow a person to identify your device as well as your general location. I'm sure you can understand why it might be better for privacy in general if everyone and every website you come across online doesn't know which town you live in.

A VPN also provides great anonymity in the sense that sites see requests from the VPN itself rather than from you. This means that it's more difficult for them to track you as an individual and build an ad profile on you. The less that companies can track you, the better protected your privacy is.

VPNs also have other non-security benefits like being able to bypass geo-restrictions. These are restrictions that only allow content to be accessed in certain locations or block content from being accessed in others. Another non-security benefit is that some ISPs throttle certain types of traffic. They might decide that video streaming is using too much bandwidth and choose to slow it down. If you use a VPN, they can't see exactly what type of traffic it is, they can only see that you're connected to a VPN.

This will make them less likely to slow down certain online activities. You can either get a VPN as an add-on to your browser or install a desktop version so that all of your non-browser traffic like games and other applications also runs through the VPN when it's active. Keep in mind that this will probably slow down the traffic so you usually don't want it running while you're playing a game. Overall Virtual Private Networks are a great way to improve your cybersecurity as well as your general browsing experience.

Another great way to improve your cybersecurity when browsing online is to use sites that are safe. Obviously, you should try to avoid sites that are illegal or appear to be scams but there's a great way of telling how safe you are when using a site. When you use your browser, the web address will usually start with either http or https. HTTP stands for HyperText Transfer Protocol and HTTPS stands for HyperText Transfer Protocol Secure.

That's right, there's a little letter in there to tell you whether or not the transfer protocol you're using is secure. To make it even easier, if websites are using HTTPS



your browser will often display a small lock next to the URL. This lock is there to tell you that the connection is secure.

Standard HTTP transmits data as plain text. This means that if someone is trying to intercept your data for malicious purposes, it's much easier for them to see what it is. This can include things like your login credentials and personal information; it's all of the information transmitted between your browser and the website you're accessing.

It also increases the risk of something like a man-in-the-middle attack which we spoke about earlier. Because the attacker can easily intercept and understand the communication, they can also easily transmit false information that looks legitimate.

HTTPS on the other hand, encrypts all of the information transmitted between your browser and the website that you're using. This means that even if an attacker manages to intercept the data, they won't be able to read it without the corresponding encryption key. This makes HTTPS much more secure against man-in-the-middle attacks and general eavesdropping. Every time you access the website there's a type of digital handshake which verifies its authenticity. Your password, username, and any other personal details are going to be much safer.

Another way to improve your cybersecurity and privacy is simply to use the internet and digital devices less. This might seem counterintuitive but professionals in the cybersecurity industry often have the least smart technology and the most basic technology in their houses. This is because it's simply easier to prevent a cyber attack on your data if your data is never in the cloud. Physical notes can't be hacked by someone living on the other side of the world and an analogue watch can't fall victim to a cyber attack. Don't feel like you always have to do everything digitally, sometimes it's okay to keep using your favorite notebook or analogue watch. If anything, it's safer.

It's also crucial to stay informed of changes and updates in the industry. Cybersecurity is a constantly changing landscape. It wasn't too long ago that people were forced to regularly change their passwords. It's generally accepted now that this is a bad practice as it leads users to making weaker passwords or improperly storing them because it's difficult to remember multiple new passwords a year.

They get what's known as "password fatigue" where they start to rely on simpler passwords or to barely change them at all. Many organizations are still forcing people to do this because they haven't stayed up to date with current cybersecurity best practices. Essentially the more up to date that both you and your software are, the better it is for your cybersecurity.

The last, and perhaps most important tip, is to be wary. Be wary of requests for personal information regardless of whether it's via email, phone calls, or online forums. Be wary of clicking links on emails if you don't know the sender.

Be wary of downloading files if you don't know the website, of using unsecured websites, of public Wi-Fi, and of granting excessive permissions to mobile apps. Be wary of anything that seems suspicious because it could compromise your cybersecurity. The less trusting you are when it comes to your cybersecurity, the better.

Chapter 17: The Hardware and Software You Need

Cybersecurity isn't just about policies and procedures, it's also about the specific hardware and software that you're using. Some devices have inherently better cybersecurity than others and the same goes for anti-malware programs, firewalls, virtual private networks, encryption software, and just about anything else.

The first thing is to make sure that the router you're picking either has a good firewall standard or can support one. Exactly which one you pick is going to depend on your network requirements including the number of devices, the type of devices, and the nature of your network's traffic.

Your router is the first line of defense and so it should also have good firewall functionalities. Without getting too complex you want to be looking for features like Stateful Packet Inspection, usually shortened to SPI, and Network Address Translation, usually shortened to NAT.

If your needs are more advanced you can also use a dedicated physical firewall device. This will have more features and generally just be better at what it does. If you're working with a smaller network, a software firewall is completely adequate. Windows, macOS, and Linux, all have pretty effective built-in firewall software. There are also third-party applications that work well. Often, it's less about which specific firewall you use and more about having one and making sure that it's properly configured.

You should spend more time making sure that it's blocking the right websites and types of traffic than you should picking which specific firewall you want to use. That said, it's worth investing a fair amount of time in both. As your first line of defense if you configure your firewall properly, all of the other cybersecurity hardware and software is just an added bonus.

Next, there's anti-virus and anti-malware software. These programs detect, prevent, and remove a variety of types of malicious software. If your needs aren't extensive, it's completely fine to use the built-in software that comes with your operating

system. Just make sure that it is actually enabled and set to regularly scan your computer. If you're at a higher risk consider using paid software from a third-party supplier like Malwarebytes, Bitdefender, ESET, or Kaspersky.

These are just some examples, depending on your region you might only be able to get from certain providers or they might offer more affordable plans. Just like a firewall, the most important thing is to have one that's installed and properly configured. They all do a reasonably good job of keeping you safe.

You also want to make sure that you're blocking as many ads as possible as these are an increasingly common angle of attack. This means that apart from your DNS which you should configure, you should also get ad blocking software. This is usually available in the form of a browser add on. You want to make sure that this software isn't just stopping you from being able to see the ads, it's actually stopping the ads from running on your device at all.

Unlike with anti-malware and firewall software, the ad blocking software that you choose to use can make a big difference. uBlock Origin has been the best at doing this for a long time and it still is despite being completely free.

Next up, you should get a VPN. Unfortunately, this is something that you're probably going to have to pay for. All of the best VPNs with the most features are paid as it costs them a fair amount of money to keep it running. They're worth paying for though as they can keep your data secure when using unsecure Wi-Fi, allow you to view content from different regions, and also stop your ISP from tracking you.

They even have the added bonus of hiding your IP address when browsing the internet. There's a lot of different companies doing this including ProtonVPN, Surfshark, Windscribe, NordVPN, ExpressVPN, and many more. They like to do lots of advertising so don't just pick whichever one your favorite youtuber advertises in their videos. Make sure that you're getting one that offers the most value for money, features, and safety.

You should also consider some intrusion detection and intrusion prevention software. This software monitors network and system activity in order to find signs of malicious behavior. Intrusion detection software will just alert you while intrusion prevention software is capable of actively blocking or preventing the threats.

Email security software can also be quite helpful. It helps to protect against phishing attacks, spam, and malware packaged into emails. It often includes things like email filtering, attachment scanning, and ways of detecting phishing attacks. A lot of email providers include some of these by default so it's worth checking which ones you already have and which ones you might need.

Also make sure that you're using a password manager. You need lots of unique and long passwords to have effective account security and it's a pain to remember all of them. Keeping them written down somewhere isn't very secure from a physical security standpoint and you want them to be available on both your mobile phone and computer.

This means you need to choose software that's secure, available on multiple platforms, and easy to use so that you don't end up making your own weak passwords. If you want all of these things there's a decent chance that you'll have to pay for them. That said, there are some reasonably effective free versions that can be found. Some specific ones to consider include Total Password, Nordpass, Bitwarden, and 1Password.

While it isn't directly cybersecurity, you probably also want to make sure that you have software for automatic backups on your computer and mobile phone. If you try and do this manually there's a decent chance that you're going to forget. It's better to just get the software for Google Drive, iCloud, OneDrive, Dropbox, or whatever other provider you want to use.

Once you've installed it you can forget about it and also have the peace of mind of knowing that all of your important data is stored somewhere safe from a ransomware or deletion attack. Also remember that if you have sensitive data, it's worth investing in some encryption software to protect it.

If you are very cybersecurity conscious it's better to use Linux or macOS for a computer as they have lower instances of viruses. Most viruses for mobile phones are also developed for Android phones, so if you are going to pick a device based off of cybersecurity, you will be better off going for a different operating system.

More important than the manufacturer, is that you're buying devices that are modern and will be supported with updates for as long as you need them to be.

If you're buying a computer, make sure that the architecture supports the latest version of the operating system it runs and that it won't be obsolete soon. If you're buying a new mobile phone, they often have very short lifespans so try to make sure that it isn't more than a year old. That way you're probably going to have security patches for its whole lifespan.

There are many other types of cybersecurity software but most of them are for cybersecurity professionals or very advanced users. As this is a book on the fundamentals of cybersecurity, we've chosen not to cover all of the advanced monitoring and management software used by those running cybersecurity departments at big companies.

If it doesn't fall under one of the listed categories and it isn't designed for cybersecurity professionals in mind then you probably don't need it. There are many companies trying to sell additional software and hardware to consumers that doesn't actually do anything other than make them think they're safer.

Lastly, and probably most importantly, make sure that all of this software that you're using is up to date. Outdated software has vulnerabilities that can be exploited and developers don't create new security patches simply for the fun of it. They create them to make sure that your devices are safe from intrusion via their software. If you're not going to remember to check that everything is updated regularly, you should probably just set it to automatically update.

Remember that none of these tools are a substitute for proper cybersecurity practices and common sense. Just like how an umbrella won't save you from getting wet if you go swimming in the ocean this software and hardware won't save you from a cyber attack if you go looking for malware. If your browsing habits regularly have you downloading software from the dark web, you should probably reconsider them.

Chapter 18: The Costs of Cybersecurity

Now that you know what's needed to improve cybersecurity in terms of hardware, software, best practices, and managing data, it's important to talk about the costs. For this section, I'll borrow my favorite economics saying, "There ain't no such thing as a free lunch." This refers to a concept in America back in the 1930s when they would offer a free lunch in order to entice customers into buying drinks.

In order for a bar to be able to offer a "free lunch" they have to charge more for their drinks. Either way you're paying, they're just structuring the costs differently. It simply means that you can't get something for nothing. Somewhere along the line, you or someone else is paying for whatever you're getting.

Everything in life has a cost, and cybersecurity is no different. It's easiest to think of cybersecurity as having three main costs types. There are the hardware costs, the software costs, and then there are the human costs.

The hardware costs can vary greatly depending on what your cybersecurity needs are. The most expensive part of cybersecurity for individuals is often the fact that you shouldn't be using out of date devices. This especially applies to mobile phones and smart devices but also to normal computers.

In order to make sure that your mobile phone or smart watch is secure, they need to have the latest security patches. Most manufactures don't run security patches for very many years after a device's release. This means that in order to get the latest software, you actually have to be buying new hardware.

The alternative to this is to buy more expensive devices like a manufacturer's top of the range smartphone. These usually gets security updates for around seven years, by which time you should be ready to upgrade your device for other reasons. Either way you're spending more money on hardware simply to make sure that you keep getting updates and have better cybersecurity.

This has also started to apply to computers, as computers that were top of the range over ten years ago can often still keep up with cheaper modern devices. CPUs made

major breakthroughs between around 2006 and 2010, after that progress has mainly been in small increments.

Because of this many of the top CPUs from around that time are still in use as budget devices. They don't have quite the same power efficiency but they're still perfectly capable of handling most modern tasks and even quite a few games. As they're not supported by Windows 11 and Windows 10 will stop receiving updates at the end of 2025, all of these computers will need to be upgraded in order to avoid becoming massive security risks.

Access devices are another hardware cost for businesses. We'll cover the cost of the access software soon, but before that you need to buy the actual access devices. There are cheaper options like hardware tokens and smart cards but these have downsides. If somebody steals an access card, they can use it to gain access to systems fairly easily. Entering a pin also isn't the most secure as it's usually only around four digits and anybody watching can pick up what the person is typing fairly easily. This is why companies often opt for more expensive access devices like biometrics. These devices eliminate the need for carrying an access card or token which also eliminates the risk of these being stolen and used by an attacker. Access devices are going to be a fairly large cost but the expensive parts of the access systems usually last for many years.

There's also the cost of cloud storage. It's a necessary part of your cybersecurity expenses so that you have backups you can restore from in the case of a ransomware or deletion attack. It also helps to prevent against permanent data loss from other things like a fire, flood, or power surge. You might be wondering why this is a hardware cost and not a software cost.

That's because when you pay for cloud storage, you're actually paying for physical data storage, it's just at a different location. If you're in a bigger company you might be able to set up your own cloud storage solution. This will cost you more initially but you could save money as you're not paying long-term monthly fees for cloud storage.

The firewall costs technically overlap both categories as firewalls can take the form of software or hardware. For more serious cybersecurity needs you're quite likely to need a physical firewall, in which case it will be a hardware cost. Then, either as well

as or instead of a physical firewall, you're going to need a software firewall. This doesn't always cost as the built-in firewalls that come with most modern operating systems are quite effective. If you need third party firewall software you can expect to pay to license it, just like any of the other software types.

The software costs are perhaps the simplest of the three categories. These costs are going to comprise of all of the different cybersecurity software that you have to pay for. If you're only doing cybersecurity for yourself or for your family, these costs are going to be relatively low.

You're quite likely to end up paying for a password manager in order to easily keep all of your passwords safe and generate secure new ones. These aren't particularly expensive and it's definitely a cost that's worth paying. Most users' passwords aren't breached because of an advanced attack.

It happens because they use very simple passwords or regularly reuse them. Password managers will even tell you if one of your passwords has been exposed by a security breach and needs to be changed. Eliminating all of these password risks with a single piece of software is one of the best value for money offerings in cybersecurity. For the average user this is probably the best cybersecurity return on investment.

Then, there's the anti-malware software. Almost all third-party antivirus software is going to charge you a licensing fee for the full version of the software. If it's just for personal use you might be able to get away with using a free version but if you're a high-risk target or if it's for business use then you're probably going to need a paid version.

If you're doing cybersecurity for a company, you're likely to also need paid SIEM software. This stands for Security Information and Event Management. This software is very important for companies as it helps you to recognize and address potential security threats and vulnerabilities. It does this by analyzing information from routers, firewalls, computers, servers, and anything else that's on your network. This allows for potential threats to be identified as soon as they happen so that you can continue without any business operations having to be disrupted.

Companies will also usually need some kind of dedicated IAM, or Identity and Access Management, software. This software manages the user accounts of

employees as well as their access rights and authentication. This is very important for ensuring that employees are accessing data that they have the correct rights for and that they're not accessing other employees' personal data or accounts without their permission.

In the case of a security breach, it can also help as it's important to trace where it came from if you want to prevent future breaches. If there are proper access management protocols in place it's much easier to see who was responsible for the breach. This software also adds a significant amount onto your total cybersecurity software costs.

Almost all of this software operates on a license basis. This means that you don't have to spend as much upfront to get your cybersecurity in order. Unfortunately, it also means that you're likely to be paying these fees for the foreseeable future. Very few companies still offer lifetime software licenses and most of them work on a monthly or yearly basis.

The good news is that cybersecurity software companies often offer discounts on their license fees. If you sign up for a whole year or if you wait for them to send you promotions, you can easily save half on your yearly fees. If you're running a company, the good news is that you can almost certainly get a discount on a bulk licensing package.

The human costs are ones that people often underestimate when it comes to cybersecurity. If you're simply doing your personal cybersecurity, it's the cost of having to train yourself and your family members to ensure that they don't compromise your cybersecurity. But when you're doing it for yourself, this can be an investment. What you put into learning about cybersecurity can easily help you land a better job or get a promotion because you're more useful at your workplace.

With companies, the human cost of cybersecurity is much, much higher. Firstly, there's the cost of paying the actual cybersecurity staff. This includes the salaries and benefits for security analysts, engineers, and managers, as well as lower-level cybersecurity staff. Salaries are a constant expense and it can often work out significantly cheaper to get a more highly qualified consultant to set up systems than to have a cybersecurity professional there at all times. That said, if it's a big enough company, you're going to need your own dedicated cybersecurity team.



Secondly, there's the cost of consultants. Just because these can sometimes work out cheaper than dedicated staff in the long run doesn't mean they're cheap. Cybersecurity consultants are highly qualified with years of studying, practical experience, and usually a number of additional certifications.

You also need to get them in reasonably often as staff, hardware, and software change with time. It's likely that you'll need to get them in at least once every three years, probably more often depending on your risks and compliance requirements.

Thirdly, there's the cost of training and certifications. Cybersecurity staff constantly have to be kept up to date with the latest developments in the field and you need to make sure that they have the right certifications. You also need to train your general staff. It doesn't help to have highly qualified cybersecurity staff if your average office worker clicks on every phishing email that they see.

For this reason, training costs can run extremely high, especially in large companies. The higher the staff turnover the worse it's going to be as every new member needs to be trained. To train them one at a time is very costly with regards to time. To hold very regular training sessions is logistically difficult and can be a pain for older staff who have already learned all of the content.

The last human cost that has to be considered for companies is that of the incident response team. If you have a smaller company, it's a good idea to have a dedicated cybersecurity with an incident response team that you use if something does go seriously wrong. These people are used to dealing with cybersecurity incidents and they're always up to date with the latest malware and techniques being used. Very large companies usually have their own dedicated incident response teams that are trained especially for situations when there's a cybersecurity breach.

Chapter 19: Identifying When Security Has Been Breached

Cybersecurity breaches aren't something that might happen, they're something that will happen. Even if you follow all of the correct steps when it comes to password creation and keeping your data safe, it's still entirely possible that a completely legitimate website or application that you use will have a breach in its security. This means that it's less a question of whether or not you will have a security breach and more a question of when it will happen.

If a hacker does manage to get into your system it's important that you discover this as quickly as possible so that you can stop them from obtaining all of your data or doing other damage. This is why it's important that you know what the symptoms of a cybersecurity breach look like. Knowing these symptoms will allow you to make a fast and accurate assessment.

First, we will look at overt breaches, you can think of these like the active attacks that we spoke about earlier in the book. Things like ransomware, denial of service attacks, and data destruction. These are usually much easier to identify than covert breaches as the hackers are actively doing something to disrupt you.

The easiest breach of all to detect is probably ransomware. This is because the attackers are attempting to contact you directly in order to extort money out of you. You can expect a large message on your screen saying something along the lines of, "All of your files have been encrypted. Don't waste your time looking for a way to decrypt them, you need our software to do so. Transfer \$200 in bitcoin to our account along with your unique key as a reference, which is RVCS-2034987.

Next, send us an email to decryptyourdata@ouremail.com and we'll send you the unique key that you need to decrypt all of your files." This is usually accompanied by a clock counting down the time that the user has to make the payment and a picture designed to scare them.

This strategy has been extremely effective and some types of ransomware are estimated to have made millions of dollars for the attackers. It's also very easy to

detect and there's almost no way that you wouldn't realize your computer has ransomware on it. If only all of the security breaches were as easy to detect!

Data destruction attacks are also usually very easy to detect. They're often accompanied by some kind of message informing users that their data has been destroyed and why it was done. If they're not accompanied by a message, you're likely to notice when you go to look for files or try to open one and it doesn't work or isn't there. They're often linked to ransomware as the software used for ransomware sometimes deletes files when the user doesn't make a payment. This will generally be done in batches in order to encourage the user to make payment. If it was done all at once, they wouldn't have bargaining power anymore.

Next, is a distributed denial of service, or DDoS attack. These can usually be detected by monitoring network traffic. For example, imagine that you have a website selling books. On average you have 20 customers a day who view the site and approximately ten of these buy books. If you wake up one day and your site has six thousand customers and only 15 book sales it's safe to say that someone is carrying out a DDoS attack. Another thing that would suggest a DDoS attack is if the traffic is coming from a large number of different countries, especially ones that your content or items for sale aren't relevant to.

There will also often be a larger amount of traffic from individual IP addresses than normal, as each one is trying to generate as much traffic as possible in order to slow down your site.

The other type of breach that you need to be able to detect is a covert breach. These breaches are significantly harder to detect as the attackers don't want you to know what they're doing. Most of these fall under the category of passive attacks that we spoke about earlier. We'll look at some symptoms of a covert breach but none of these exist in isolation. You need to remember that having one or even multiple of these system does not necessarily guarantee that you have a security breach.

But, if a device that has been working fine suddenly develops multiple of these symptoms, it's likely that there is a security breach. It's also relevant to try and remember what happened before the symptoms started showing. To give an example, if your device is slower after your operating system has just updated, it's quite unlikely that it's the result of a security breach. If your device is slower the day

after your friend gave you a copy of a pirated game, you've probably had a security breach.

The first sign that your security may have been breached is often your device running slower than before. This is because the malware is usually running in the background and often transmitting data. This will slow down both your actual device and your internet connection. Like before, it's important to keep perspective on this. If you're unsure why your device is slowing down, opening task manager using Cntrl + Shift + Esc is a good way to check. You can sort by CPU or memory usage to see why your computer is running slower.

It's normal for something like your browser to take up a lot of RAM, but if you see a program you don't recognize using lots of resources, it's a good idea to google the name and see what comes up. Sometimes standard Windows processes will use higher resources but this shouldn't go on for days or weeks. It should be an occasional occurrence before or after Windows updates or during antivirus scans.

If your task manager doesn't run at all when you try to open it, this can also be a sign of malware. Some types of malware intentionally stop task manager from running so that users cannot check their resource usage or figure out what processes are running. Some types also mimic common Windows processes and copy their names so that users assume they aren't harmful. If there is a recurring Windows process slowing down an otherwise fast computer it's a good idea to run a malware scan just in case it's malicious software.

The same goes for the registry editor not running in Windows.

Another symptom of malware is that your device has high latency. This means that there is a long delay between you searching for something online and the request going through, or you clicking send on an email and it actually being sent. If this is happening on all of your devices then it's quite likely that you simply have a slow internet connection. If it's happening on only one device then it's probably time to investigate further.

The same goes for videos loading slowly and frequently buffering. If a device that was working fine before is suddenly loading videos slowly on the same internet connection, it could be a sign of malware. If you have a metered internet connection



you can also expect your data usage to go up as the malware is sending data from your computer.

One common sign of a security breach is if friends, family, or coworkers are receiving communication from you that you didn't send. For example, coworkers are receiving spam emails coming from your email account or friends have been receiving strange WhatsApp messages with download links in them.

If it's a laptop or a mobile device that's been breached it's quite common for them to feel hotter. This is because they're constantly working even when you're not doing much on them, which causes the CPU to run hotter.

This will also result in your battery going flat faster as more power is being used for other resources. Higher resource usage and devices running hotter can also lead to them crashing. If you regularly see the blue screen of death on your Windows computer that says, "Your PC ran into a problem and needs to restart. We're just collecting some error info, and then we'll restart for you" this could be a sign of malware. If it isn't there's probably another issue on your PC that you need to check out.

It's also very common for malware to result in popups in unexpected places. If you're browsing the web without an ad blocker seeing a few adverts is normal. However, if you start seeing popups on your home screen or while typing up a Word document, it's safe to say that there's some kind of malware on your device.

A common vector of attack is for malware to change your browser settings. If you see your default browser has been changed to one that you don't recognize, your homepage is different, or your browser seems to be using a different search engine, you know that something is wrong. The goal of these browser attacks is to redirect you to other malicious sites that can then be used to scam you or steal your personal information.

Sometimes you might not even have to detect the breach yourself. If your account has been breached that you use to login to a website or application, there's a decent chance that they'll find it before you do. They'll probably send an email informing you that you need to change your password as a result of a security breach. If you see one of these emails, remember to verify it first to avoid giving attackers your personal details.

As you can see from all of this, there's generally no one symptom that means you've definitely had a security breach. Aside from something very obvious like a ransomware attack, it can be quite difficult to detect. It's often just that your computer feels a bit off, like it isn't working quite how it used to. Settings might have changed that you don't remember changing, or it might suddenly be restarting for no apparent reason.

You know your computer, how fast it loads, how often it's set to install updates, and how much your videos buffer. Ultimately, you're the best person to be able to pick up if there's covert malware installed on your device. Because of this you should always stay alert and aware of any changes that aren't a direct result of a legitimate program installation or update.

Chapter 20: Recovering From a Breach of Security

So, the worst-case scenario happened. You realized that you've had a breach of security. Perhaps you booted up your computer and there was a ransomware message. Maybe your antivirus program informed you that it found a threat and it's been quarantined. Or, you might have noticed suspicious popups on your desktop where adverts shouldn't be able to show up.

What do you do now? How do you recover from the security breach?

Before we start there's an important disclaimer that you need to know. If you can, it's always best to call a cybersecurity professional. If you're running a company, you probably have the resources to be able to afford one. If there was an irregularity with your bookkeeping, you'd call an accountant. If your car broke down, you'd call a mechanic. If your house was broken into and everything was stolen, you'd call the police.

If you've had a cybersecurity breach, you should call a cybersecurity professional. They know more about cybersecurity than can be learned from a few google searches and they've probably seen problems just like yours one hundred times before. They can diagnose and deal with the problem quicker and with less adverse consequences than you can. That said, you might not know any or not be able to afford their services.

The first thing to note is that it's a lot easier to recover from a security breach if you were following the guidelines laid out in this book about how to protect yourself. If you already have your data backed up, anti-malware software installed, and a proper firewall on your network, it's going to be a lot easier to recover from the breach.

At this point you might be angry, frustrated, terrified, anxious, or all of the above. It's important to take a deep breath and stay calm. You need to act logically and not panic but also quickly. With cyber attacks your biggest enemy can often be time. The longer you leave malware unchecked for, the faster it can spread. If it's ransomware

you're often working against a timer that will delete your data if you take too long. So, what do you do?

The first step is to determine as much as you can about the breach. You need to think of yourself like a doctor, healing a disease. The more you understand about the disease, the easier it is to figure out what needs to be done. This means figuring out as much as you can about the specific symptoms of your cybersecurity breach.

What made you notice that there was a breach in the first place? You should take the time to try and figure it out, but ideally be quick about it. A cybersecurity breach isn't something that you can leave until tomorrow in order to figure out what went wrong.

Let's say that you identified the breach by a message on your screen informing you that your computer has been encrypted and you need to pay a certain amount of money in order to get your data back.

From the symptoms described, you might remember this as a ransomware attack. In this case, the first thing you would need to do is disconnect your PC from the internet and your local network. The fastest way to do this is either to pull out your LAN cable or to disconnect your Wi-Fi. This will help to stop the ransomware from spreading to other devices or from downloading additional malware to your device.

The next step is to identify the specific variant of ransomware that you have. The easiest way to do this is to put your symptoms or some of the text found in the ransomware message into Google. Usually, the specific email addresses and bitcoin wallets provided for the ransomware change most of the message and way that the software works remains the same. You should pretty quickly find results detailing the type of malware that you have and whether or not the data can be recovered.

Now, hopefully you took the advice given earlier in this book and backed up all of your data, software keys, etc. If this is the case, the easiest way to deal with the ransomware would be to simply reformat your hard drive (or drives). A clean reformat along with a new installation of your operating system is almost always enough to remove any malware located on your computer.

But maybe it wasn't a ransomware attack, maybe it was something else like a phishing attack. Perhaps you were tired one night and you accidentally put your details into a phishing email. When you woke up the next morning, you realized your

terrible mistake. Now you need to figure out what information the attackers have and what they're capable of doing with it.

In this case, the first step will be to go back to the email that they sent you and figure out what details it was that they were trying to obtain. What they were trying to get and what you gave them will determine how you need to act. So, let's say that they were impersonating your bank and you gave them the login details for your online banking.

If this was the case, you immediately need to notify your bank. Not using the details on the phishing email, but using the real helpline number for your bank. Once they're aware they can put a freeze on the account, stopping any outgoing transactions until they've sorted out the rest of the details. It might be scary doing this but it's much better to do it and feel silly for worrying about it than it is to not do it and have all of the money taken out of your account.

The next step would be to change your password. This applies to any account information that you might have had stolen. You need to change the password for that account as well as any other accounts that you might have used the same password for. As you can tell, this is why it's important to always use unique passwords! You also need to think about what other accounts they might have been able to gain access to. For example, if your email password was leaked, the attackers could gain access to almost any account linked to your email by resetting the passwords for those accounts.

It's generally a good idea to do your best to inform everyone who might be affected that you've fallen victim to a phishing attempt. That way, if they notice suspicious information on any of your accounts, they can notify you. Next, you should do a full malware scan on the device where it happened. You might not have noticed but it's possible that the phishing email downloaded malware, or redirected you to a site where it was downloaded.

As you can tell, different types of attacks need to be dealt with in different ways, and outlining all of the ways in which each unique attack type and incident needs to be dealt with could be its own whole book. For this reason, the following steps are some general steps that you can take if there's a breach of security and you're not quite sure exactly what went wrong.

If there's a breach of security in the form of malware installed on a computer like a virus, worm, trojan, ransomware, or keylogger, the first thing to do is to disconnect it from the local network and the internet. Usually, the quickest and easiest way to do this is by unplugging the LAN cable or switching off the Wi-Fi. This helps to ensure that the malware doesn't spread to other devices in your local network or over the internet. The next step is to run an anti-malware scan in order to find and remove all of the malware on your computer.

If this doesn't work because the malware is stopping the program from running or you're worried that it hasn't all been removed, the next step is to reformat the hard drive. Keep in mind that reformatting a hard drive will remove all of the data on it, so it's not an ideal first step. It is however a highly successful way of removing all malware on a device.

Ideally your data will be backed up along with all of the keys you need to reinstall your software, like we discussed in the previous chapter on how to protect your data. If this is the case, you should be able to reformat everything, reinstall all of your programs, and redownload your data from the backup with minimal data or time loss. If your data isn't backed up, you'll have to be much more careful about reformatting the drive and if an anti-malware program can't fix your issue, you'll need to consult a cybersecurity professional.

If you feel that one of your accounts like your bank account, online shopping account, or social media account has had its security breach and the attackers have access to your login details, the first thing you need to do is notify the company that the account is with. Hopefully they'll be able to put a freeze on the account and stop any unauthorized activity. The next step, and you don't have to wait for the first one to do this, is to change your password. You should change the password for the account that had its login details compromised as well as any other accounts that you may have reused that password for.

I'm sure that by now you're understanding why unique passwords is so important. You should also change the passwords for any other accounts that the attackers may have been able to gain access to using your compromised account. For example, if your email account password has been leaked, attackers could use this information to request password resets on your other accounts. The password reset information will be sent to your email account and they can then gain access to all of your other



accounts via your email. If you're unsure if a specific account has been compromised, it's better to simply change the password in case it has.

Hopefully this information allowed you to recover from a security breach and you didn't lose too much data or money in the process. But just recovering isn't enough, it's important to understand why it happened and what can be done to prevent it in the future.

Chapter 21: Making Sure It Doesn't Happen Again

Now that you've recovered from your security breach, it's time to make sure that it can never happen again. Hopefully, you've already determined what the breach was in the previous chapter as well as what caused it. It's very possible that the attackers will be aware that they succeeded against you the first time and they might be looking to pull off a second attack as they believe you to be an easy target. How you make sure a cybersecurity breach doesn't happen again is largely dependent on what caused it in the first place. Let's look at some common ways that the attacker may have gained entry to your system and what you can do to prevent it from happening next time.

The most common way that attackers gain entry is via people. I'm sure as you've learned more about cybersecurity you've come to learn that people are the weakest link. It's likely that an attacker gained entrance via a phishing email, text message, or voice call. If this happened, you need to look at who gave them this access. It's unlikely that it was you as you're reading this book, so you probably have a reasonably good idea of what phishing is and to be alert for it. You already know to check for inconsistencies in spelling, being redirected to fake websites, and emails asking you for your details that don't seem trustworthy. This means that it was probably a family or staff member.

They need to be educated on cybersecurity so that they understand what the risks are and how to avoid it from happening again. If you don't have a cybersecurity expert to explain it to them and you don't want to do it yourself, you could even direct them to this book. Even if all they pay attention to is the chapter on different cybersecurity attacks, it should still be enough to prevent them from falling victim to more simple types of social engineering.

You also need to look at who has access to your computer systems and consider restricting access. Children can be given accounts without administrative permissions which are much less likely to expose the whole computer if they click on something that they shouldn't. If a person can't be trusted not to fall victim to a basic

phishing attempt, then it might be easier and safer if they don't have access to administrative permissions. This also applies if you're in an organization like a school or business. The less permissions that users with limited computer understanding have, the less chance they have of accidentally breaking something on the system.

You should also seriously consider making sure that you have a firewall on your router and that the firewall is optimized to protect you from cyber-attacks. If you're in an organization or large household you could also consider blocking certain sites that are leaving you vulnerable to attacks. For example, if someone keeps downloading malware because they're trying to pirate games, you could consider implementing a network wide ban on piracy sites.

There are many lists that can be used without you having to individually list and block problematic websites. It's better if most malicious traffic can be blocked before it even reaches the computers on the network. You can also look at implementing a private DNS on your router in order to block ads. This means that people on the network will see less ads and therefore click on less ads. As we already know, ads are an increasingly popular attack vector, so the less people see the less chance they have of being redirected to a shady download link.

It's also vital that you make sure to check who you're receiving data from. If friends, family, or employees are regularly exchanging flash drives, then your computer is at risk of being infected by anything they have on theirs. Spreading through flash drives is a common vector of attack and you need to be sure that everyone is aware of this. This is one of the reasons why anti-virus and anti-malware software is so important. Ideally your software should be scanning every flash drive that you put into your computer.

As you can tell, for anything malicious that does reach the computers, it's important that you have anti-virus and anti-malware software installed. If the built in software that came with your operating system didn't detect the malware the first time, it's probably a good idea to get third-party anti-malware and anti-virus software. This also applies if you feel you're high-risk or if you frequently need to use less than reputable sites for your job or anything else.

It's better to avoid dangerous sites altogether, but if you do decide to frequent the dark web or other suspect parts of the internet it's best to make sure that you're as protected as possible. You can think of it like going for a walk late at night in a bad part of town. It's an activity that's better avoided altogether but if you do make a habit of it then you should probably be carrying a taser. In this case, your taser is anti-malware software.

In order to prevent future cybersecurity attacks it's also important that everything is up to date. Outdated hardware and software are common ways that attackers gain entry to a system. If you're using devices that are no longer receiving software updates then you should consider changing them for newer versions. With many devices, smart devices and mobile phones, manufacturers only continue to update the software with security patches for a few years. If you can't afford to upgrade them then it might be worth looking at switching the operating system running on your devices. For example, if Windows 10 is no longer receiving updates on a computer and it can't upgrade to Windows 11, you can instead switch to a version of Linux that will continue to receive security updates.

If the attacker gained access to an account rather than a physical computer, you need to look at what gave them access to this account. If it was a weak password then it's time to start using a password manager to generate and manage all of your passwords. Also, consider using two factor authentication in order to ensure that even if your login details are leaked, hackers still can't easily access your account.

Another important thing is to make sure that all important data is both encrypted and backed up. This ensures that if it's held ransom or needs to be deleted in order to reformat a drive, it's easy to get all of the information back and continue without any issues.

If you're running a business or are a high risk target it's also important to consult with cybersecurity professionals. They will be able to examine your specific circumstances and give advice that will help improve it in a manner that is both practically effective and cost effective. They can also develop systems for employee training and data handling that can then be taken over by less skilled (and more affordable) people at a later stage. Sticking to these policies can greatly reduce your risks of a repeat occurrence. It could also be worth sending whoever handles your IT for advanced training on cybersecurity so that they're better equipped for future



incidents. If you're in this position it's also a good idea to stay as up to date on the latest cybersecurity attacks as possible. If you know how attackers are currently exploiting systems then you will know what the best steps are to avoid these attacks.

As you can tell there's a wide range of ways in which cybersecurity incidents need to be handled and how you prevent a future one is largely based on your specific weaknesses and vulnerabilities.

That said, if you follow all of the tips and tricks in this book, you should be well equipped both to recover from a cybersecurity attack if it does happen and to prevent it from ever happening again. The most important thing is for you, and the people around you, to be educated on cybersecurity attacks so that you recognize one when you see it.

Chapter 22: Important Business Cybersecurity Breaches

There have been many breaches of different businesses' cybersecurity but this chapter will focus on some of the most infamous ones. These have also been picked specifically because there are cybersecurity lessons that can be learnt from them. They either prompted changes within the industry or served as a lesson to others of what would happen if they didn't follow the strict policies and procedures that sometimes seemed unnecessary.

Perhaps the biggest business cybersecurity breach of all time was Yahoo. Yahoo is an American company that provides a variety of online services including sports news, email, financial data, general news, and even a search engine. The company was established in 1994 and was one of the largest internet companies before losing a significant amount of market share to Google and Facebook. The breaches occurred significantly after Yahoo was at the height of its popularity but a huge number of users were still affected. There were two breaches and an estimated three billion accounts were affected by them.

The first breach occurred in August of 2013 when cyber criminals managed to gain unauthorized access to Yahoo's user data. This included things like their names, telephone numbers, email addresses, and hashed passwords. A hashed password is a password that's been turned into a short string of letters and/or numbers so that it can't easily be used following a data breach. If the passwords weren't hashed, the breach could have been substantially worse. This personal data that was leaked in the security breach was that of an estimated one billion users.

At least, that's what Yahoo initially thought. They later believed that all three billion users had had their data stolen. The breach was only discovered when Andrew Komarov, the CEO of a cybersecurity firm, saw the information for sale on the dark web. This was already in 2015, two years after the data had been stolen. Komarov then alerted the relevant authorities as the names and personal details of high-ranking politicians and military officers were found to be on this list.



The second breach occurred in 2014 when hackers managed to obtain more of the same information as before including names, dates of birth, telephone numbers, and security questions of over 500 million users. Cybersecurity experts and authorities were concerned that this information could be used to obtain login details for other accounts by using phishing or other methods of attack. Imagine that you receive a phishing email saying “Dear Yahoo user, please click this link to verify your account details”.

It simply isn't very convincing. Now, imagine that you receive a phishing email that refers to you by name and asks you to verify that a specific cellphone number is indeed your number as well as your password. Because they know your details, you'd be much more likely to believe them. They could even set you up for it by notifying you of a data breach via text message and then sending you an email asking you to enter your old password and make a new one to keep your account secure. All of this would be possible because they had your personal information.

Yahoo failed to disclose these breaches of security until late 2016. The fact that they did not inform users or shareholders of the breach promptly raised serious security concerns around the company. The company gained a reputation for having lax security measures and for not taking proper care of customers' data. As a result, the company ended up having to pay out around \$50 million via class action lawsuits and also lost a lot of value.

There are a number of lessons to be learned from this breach. It highlighted the importance of disclosing information to users in a timely manner so that they know what's actually going on with their account. It also showed organizations how big of a target data is and that they need to better protect user data and implement more effective policies. Lastly, it laid the groundwork for international cooperation in addressing cybersecurity threats. As the data for many important government and military officials in different countries was leaked, it showed authorities the importance of being able to cooperate with one another to tackle cybercrime.

Another major cybersecurity incident was the security breach of Equifax Inc. Equifax is a credit reporting agency based in America and it is one of the largest three in the world. It collects the data of over 800 million individual consumers as well as 88 million businesses, located all over the world. It's a massive company with over US\$5 billion in annual revenue as well as over 14 000 employees located in 24

different countries. I'm sure you're beginning to get an understanding of just how big this company is and how disastrous a security breach could be. So, what actually happened?

In 2017, what's now considered one of the largest and most impactful data breaches in history occurred. Somewhere between May and late July of 2017, the data of around 147 million people in the United States that use Equifax was leaked. This data was highly sensitive as it contained physical addresses, names, Social Security numbers, birthdates, and even some driver's license numbers. Equifax discovered that a breach had occurred on the 29th of July 2017 and informed the public on the 7th of September 2017. As you can tell, the time between discovery and disclosure had already gotten a lot shorter following the infamous Yahoo security breach. Most people still considered this too long of a delay, and just like Yahoo, Equifax faced significant backlash for their delayed reporting on the breach.

Attackers managed to exploit a vulnerability located in the complaint web portal. This web portal used open-source software that had a known vulnerability. The vulnerability was discovered and patched but Equifax failed to do so in time. This also caused criticism as many people felt that Equifax could have prevented the data breach entirely if they were simply more security focused and had more effective internal policies for preventing this sort of issue.

In addition to this during the delay between the breach being discovered and the breach being made public, many of the company's top executives sold shares that they held in the company. This caused the public to believe that they had done so as they knew the share price would drop once the information was publicized. Despite this, only one lower-level executive was ever charged with insider trading.

When the data was leaked there was a major risk of identity theft as the hackers had access to enough personal information like Social Security numbers to imitate various individuals. Because of how sensitive this data was and how badly they handled the incident, Equifax faced numerous lawsuits as a result of the breach. These were from people whose information was leaked as well as various US government agencies whose regulations they had breached.

Equifax is considered a cautionary tale of what can happen when cybersecurity protocols are ignored, and top executives become lazy with data protection. It



highlighted the need to deal with known vulnerabilities as quickly as possible in order to protect consumer data. As it came not long after the massive Yahoo breach, it also prompted discussions on the responsibility of companies to protect user data. This led to increased regulations around company cybersecurity standards and how user data is handled.

The third and last major cybersecurity breach that we will be looking at is that of Adult Friend Finder. This is an online company that primarily deals in adult entertainment, a type of adult social media, as well as online dating. It originally started as another site called FriendFinder where people could find friends online.

When the founders discovered that users had been using the website to find sexual partners, they released a spinoff named AdultFriendFinder. This also branched into various other niche dating sites for users of specific religious orientations or other preferences.

As you can imagine, a lot of people wanted the fact that they were using this site kept secret, and the data breach represented a massive loss of privacy. In 2016 the information of around 412 million accounts was leaked. This included personal data including email addresses, usernames, and other more compromising information about users' adult preferences and activities. The attackers exploited a breach in the site's password storing practices. They used outdated and insecure encryption methods that should have been phased out years before.

The site encouraged users to change their passwords once they discovered the breach and tightened up on their security measures. This was a massive breach and largely caused by negligence on the company's part. This doesn't, however, make it unique. What makes it unique is the type of site that it was. With the other breaches users were upset that their date had been leaked but with AdultFriendFinder many users didn't even want people finding out that they had been using the site at all.

There were concerns about the embarrassment for users as well as potential blackmail or extortion by attackers who could expose their personal information. Many users were terrified that their friends or family would find out that they had been using the site and they would face public shame.

This case only further reinforced how important it was for companies to be taking care of their user data. This is regardless of the industry that they operate in and the

more sensitive the nature of the site, the more upset users will be if their information is leaked.

As you can tell all of these breaches happened around a similar time and they forced some necessary changes in the industry. Companies were shown how important it is to protect user data as well as how big of a target they are for cybersecurity attacks. This caused them to place a larger emphasis on cybersecurity as a whole. They developed better policies for incident response and devoted larger parts of their budgets to ensuring that their cybersecurity was improved. As the news articles on these breaches reached the general public, it caused them to begin to take a greater interest in cybersecurity. This meant that there were also legislation and regulatory changes prompted by these incidents.

For example, the European Union implemented what's known as the General Data Protection Regulation or GDPR in 2018. This imposed strict requirements on how personal data should be protected. While these breaches were obviously unfortunate, they also taught us a lot about cybersecurity and led to many great changes being made in legislation as well as industry practices.

Chapter 23: Important Individual Cybersecurity Breaches

In the same way that some of the major cybersecurity breaches forced companies to place a bigger emphasis on cybersecurity, there were others that forced individuals to do the same. In this chapter we'll take a look at some of the majorly publicized individual cybersecurity breaches and how they forced the public as a whole to adopt better cybersecurity standards.

Perhaps the most majorly publicized incident was the 2014 leak of various celebrities' pictures and videos. These were not from movies or series that the actors starred in, rather it was the contents of their private mobile device's galleries. These pictures were obtained via iCloud, an Apple service that the celebrities used to backup their devices. It was initially believed that this was done via an exploit in the Find My iPhone service or some vulnerability in the iCloud platform itself. If this was the case, it would have been in the previous chapter and the private data of millions of users would probably have been leaked.

Instead, it was the result of incredibly targeted attacks on specific individuals. This was obtained using methods like phishing but also brute forcing and simply guessing their passwords until they got one right. With a normal user this would be much less of an issue, but with celebrities there were many more people invested in gaining access to their accounts. In addition to this, a lot more information on celebrities is publicly available.

They often do interviews or even films that show their spouses, dogs, where they grew up, and what hobbies they have. This makes targeted attacks much more effective as so much of their information is publicly available. In addition to this, certain attacker did things like create an account named "apple privacy security", which they used to ask celebrities for their security information as a form of phishing.

These pictures and videos, often intimate ones, were then sold on online forums for Bitcoin or were otherwise distributed. Instead of it being a single attacker it was a

number of attackers, some who faced sentencing and some who were never found. The longest sentence given out for the crime was only 34 months, with the shortest being 8 months. The hackers were made to pay almost nothing in restitutions.

Most of these were not working together, they simply independently attacked various celebrities. The information was then aggregated into one place as it was posted online, making it seem like one massive breach of security rather than a collection of successful phishing attempts.

Cybersecurity professionals were quick to point out that the hacks could have been prevented had the celebrities used two factor authentication or been more wary of phishing attempts. This massively publicized breach of security helped to show the public just how important it is to make sure that your accounts are secure. It also showed the general public some of the dangers of using cloud storage and that they should think about what kind of content they have stored online. After the breach there was an increase in people using two factor authentication and more people realized that they needed to be using stronger passwords.

Another piece of malware that greatly affected the general public was malware known as WannaCry. This was an infamous ransomware attack that encrypted user's files and demanded three hundred dollars in bitcoin to release them. The exploit it used to do this was known as EternalBlue, an exploit actually developed by the United States National Security Agency, commonly referred to as the NSA. This attack happened worldwide in May 2017 and targeted computers running Microsoft Windows.

Windows had actually already released a patch for the exploit a month earlier, but the ransomware was still effective against computers that had not yet been patched or were still using outdated versions of Windows that stopped receiving support. The attack affected over 300 000 computers in the space of only four days. Estimates for how much the attack ended up costing range from millions to billions of US dollars.

As the exploit was developed by the NSA, many people blamed the US government for the attack, stating that if they had disclosed the vulnerability when they discovered it, people could have patched their computers in time. Many people also felt that the intelligence agency's tendency to keep exploits secret so that they could use them was against the best interest of the public. Microsoft themselves accused

governments across the world of hoarding vulnerabilities instead of disclosing them so that they could be patched. While this attack had a huge impact on business cybersecurity, it also caused the general public to take note.

Many people began to understand the need for security patches and were less likely to ignore them. Ignoring a Windows update when you think it's for something you don't need is one thing, ignoring a Windows update when it could be a security patch that saves you from ransomware is another. WannaCry helped average computer users to understand that security patches weren't something you should delay or ignore and that running out of date software has some massive risks.

Another cybersecurity breach that massively infected individuals was the 2020 Twitter account hijacking. On the 15th of July 2020, 130 high-profile twitter accounts were taken over by attackers. They used social engineering on Twitter employees in order to gain access to administrative tools that allowed them to alter accounts and post tweets directly. The scam tweets asked the general public on twitter to send bitcoin to a specific cryptocurrency wallet. They promised that the money would be doubled and returned to them. If this sounds too good to be true, that's because it obviously was.

Only 45 accounts out of the 130 compromised actually tweeted the message, but these were incredibly high profile. Some of them included major cryptocurrency accounts like Coinbase, CoinDesk, and Binance, as well as famous people like Elon Musk, Bill Gates, Joe Biden, and Barrack Obama. They also obtained access to the accounts of other major companies like Apple. Twitter removed the tweets within minutes but this was still enough time for over one hundred and ten thousand dollars in bitcoin to be deposited into one of the wallets.

This is often considered to be one of the worst cybersecurity hacks of all time. This partially because of the high-profile accounts that the attackers managed to gain access to, but also because of the fact that it was done via employees. It caused a greater awareness of the dangers of social engineering as well as phishing.

The last attack that majorly affected the public that we'll be looking at is known as the Ashley Madison data breach. Ashley Madison was a website designed for people have extramarital affairs. The website's slogan was "Life is short, have an affair" and it was essentially a dating website for married individuals.

The website was widely criticized for making money off of damaging relationships and in July 2015, a group calling themselves “The Impact Team” announced that they had stolen the website’s user data. They told the website to shut down and if it did not that they would leak all of the users’ personal information.

As proof that they had the information, the attackers leaked the personal information of 2500 users. Despite this, the company denied that their records were at all insecure. They continued to operate while user fear mounted that they would have their personal information leaked and they would be publicly shamed. These fears were in part because the website didn’t delete the personal details of users once their accounts were deleted. In order to have their records deleted, they had to pay the company.

It turns out that they were right to be worried, as in August over 60 gigabytes of company data was public released. This included user data like their names, home addresses, search history, and credit card records. Even the users who had paid to have their account information deleted had it publicized as Ashley Madison had failed to honor their end of the agreement.

Some might find this ironic as the users of the site failed to honor their end of their marriage agreements. Unsurprisingly the holding company of the site, Avid Life Media, found themselves the target of multiple lawsuits. In 2017 they agreed to settle these for eleven point two million US dollars. As the information continued to be leaked both on the dark web and normal internet, people began to comb through the data. One thousand two hundred Saudi Arabian email addresses were found to be signed up to Ashley Madison.

This was highly problematic as adultery is punishable by death in Saudi Arabia. In addition to this there were many other accounts linked to high profile figures either in religious communities or in the government. There were even a few suicides linked to the security breach as certain people felt that it would ruin their public image and life. To make matters worse it was found that most of the accounts that men had been chatting with on the site were in fact fake bots, designed to make them spend more money on the website.

The Ashley Madison leak showed people that it wasn’t just important to have secure passwords and antiviruses on their devices. It showed them that it was also



important who they trusted with their data. Shady sites could easily leak their private information, leading to both financial and personal loss. This example served as an important lesson on why users need to carefully manage their privacy and on just how public the internet can be.

While all of these cybersecurity attacks were damaging for the individuals affected, they all furthered the average person's understanding of cybersecurity. As a direct result of them many more people are now adhering to basic cybersecurity principles.

They've caused users to be more careful with their privacy, more diligent with their passwords, and more wary of phishing attempts. Ultimately there's a lesson in all of these incidents and they're a good showcase for why some of the cybersecurity principles that we adhere to are so important.

Chapter 24: The Origins of Cybersecurity

If you've made it to this part in the book you've already learned a lot about cybersecurity. You know more about the different types of attacks that are commonly used, how they carry out those attacks, and even how you can protect yourself from them. You should have a deeper understanding of why different types of hackers carry out the attacks that they do and what they're looking to gain.

You might even have begun to think that there could be a future for you in cybersecurity. Perhaps you want to pursue it as a career and become a white hat hacker, earning a living from helping companies find the vulnerabilities in their systems and advising them on how to patch these holes.

But where did cybersecurity actually come from?

When the first computers were developed in the 1940s, they were enormous. One computer would fill an entire room and they weren't networked to anything. The focus was mainly on seeing how they could be improved and developing new technology. Most people couldn't understand what was on computers and nobody was thinking about trying to exploit vulnerabilities.

In the 1960s time-sharing was developed. At the most basic level this allowed multiple users access to a computer simultaneously, similarly to how schools and businesses often run servers with multiple terminals. As multiple users were on one computer it started to become evident that it would be important to be able to protect the users' data. This was still limited to academic research and discussion however.

In the 1970s when they began to link computers together into networks people began to grow more concerned about the security risks. These early networks laid the groundwork for the modern internet, which took off in the 1980s. Despite this, it was not yet easy to navigate or use. Most people didn't have a computer and the first computer viruses which started being developed around this time were still relatively easy to detect.

January 1 1983 is generally accepted as the official birthday of the internet. It was soon after this that some of the earliest and most serious threats to computers security began to develop. As computers were still developing, dedicated antivirus programs did not yet exist. If a new threat was detected, someone would write a program to remove that specific threat. The antivirus programs that were developed were extremely limited and updated infrequently.

In 1988 the first computer majorly destructive computer worm, called the Morris worm, was distributed via the internet. This caught mainstream media attention and resulted in the first felony conviction in the US under the newly developed Computer Fraud and Abuse Act. The worm itself ended up being more harmful than even Morris himself expected. The initial plan was for it to check whether or not a computer was already infected before it copied itself. Morris was however worried that this would be easy to counter as system administrators might simply tell the computers to report a false positive.

For this reason, he told the worm to copy itself 14% of the time. What he didn't anticipate happening was that already infected computers would be infected over and over again. As a result of this they began to slow down until they were unusable and the computers would crash. This truly showed both system administrators and the general public just how dangerous attacks could be and how important cybersecurity was.

Soon after many major Antivirus companies were founded. These included some companies which you might still hear about today like McAfee, Kaspersky, and Avast. This was a beginning of a new era, not because of the newly founded antivirus companies, but rather because the internet was finally gaining widespread use. As the general public began using the internet there was a massive increase in cyber threats. Suddenly viruses, worms, and cybercrime were widespread.

This meant that everyday people now had a need for antivirus programs. The industry began to boom and in 1991 the first version of Norton Antivirus was released in the United States. In the very same year the first version of AVG (Antivirus Guard) was released in the Czech Republic. Only a few years later other well-known companies like Kaspersky, and Bitdefender were founded.

As suggested by the name, Windows 95 was released in 1995, and an Avast employee wrote the first antivirus program for the operating system. During this period new viruses and cyber attacks were launched regularly and it soon became common knowledge that you needed an antivirus program on your computer. People also began to be smarter about where or from who they got their programs and files.

The early 2000s was a time when the cybersecurity industry was in full swing. Both governments and organizations began developing cybersecurity practices as well as common standards for the industry. The total number of malware samples in antivirus databases exploded and as we moved into the 2010s cyber threats became increasingly more sophisticated.

Large-scale data breaches as well as ransomware attacks became significantly more commonplace. This caused companies to invest in better cybersecurity measures and the profession continued to grow both in importance and popularity. When buying a new laptop or computer users would often start by installing an antivirus before getting the other necessary software. It was simply what you did when you got a new device.

When Windows released Windows 8, they bundled their own free antivirus protection into the software. This was, and still is, called Windows Defender. The version that came with Windows 8 wasn't the initial release, it was an optional download for Windows XP and later Vista and 7. But only when Windows 8 was released did it become its own full standalone antivirus. Since Windows 8 all Windows computers come with Windows Defender installed and automatically enabled. It will only be disabled if the user decides to download a third-party antivirus app. In 2012 Apple released their own software called Gatekeeper. This is designed to help users only run trusted and security checked software on their Macs.

As time went on Windows Defender became an increasingly competent antivirus application. While it is not necessarily the best in the field, when considering the fact that it's free with Windows, it's a very decent offering. As other free alternatives have become increasingly intrusive, constantly showing pop-ups and asking users to upgrade to premium versions, Windows Defender has only increased in popularity. With later versions of Windows, it has become deeply integrated into the operating system and many people use it simply because it is too much effort to remove.

Some people say that this is to prevent viruses from removing the program while others claim that it's Microsoft's way of pushing their program onto users.

Whatever the reason, it looks like built-in antiviruses and other security software is here to stay. This means that your average user's computer is possibly more secure than ever before. In addition to this, one of the most recent advances in the field is the use of machine learning and artificial intelligence.

As artificial intelligence like ChatGPT becomes a part of our everyday lives, it also becomes a massive part of cybersecurity. Professionals in the field are using these tools to train software to respond to threats in real time. Instead of only being able to respond to exist threats that they can detect, artificial intelligence will learn to anticipate new threats and detect malware in real time.

You can also be sure that there are people looking into training artificial intelligence to create new malware. With a big enough database of information on quality exploits and system vulnerabilities, AI could begin to generate malware faster than humans ever could.

Not just this, but it could also be used to create phishing emails, and much more. We won't go too far into detail here as we'll cover artificial intelligence's role in the future of cybersecurity a bit further on.

Another problem is, as security that comes installed on devices by default gets better, users grow more complacent. With the Internet of Things gaining popularity, people have more smart devices than ever before. Manufacturers frequently stop releasing updates to devices and homes have more unsecured and obsolete devices waiting to be exploited than they ever did.

What does this mean?

The history of cybersecurity isn't over, it's only just being written. As you'll see in some of the later chapters, cybersecurity as a field is here to stay.

Chapter 25: The First Cybersecurity Breaches

Now that you know a bit more about the history of cybersecurity, we're going to look a bit more in depth at some of the first attacks. These attacks are important because they showed us why cybersecurity was so important and what it could become as a field.

The very first example of malware ever was called Creeper and it was designed in 1971. Creeper wasn't actually malicious, rather it was created as an experiment and proof of concept. It was a self-replicating worm that would move between computers. It displayed the message "I'M THE CREEPER : CATCH ME IF YOU CAN" and apart from that had no adverse effect on them. Later in 1972, Reaper was designed as the first example of anti-virus software. Its goal was to move across the infected computers and "catch" Creeper by deleting it.

While this wasn't a true cybersecurity attack, it was proof that one could exist. It helped people to understand how software could potentially be harmful and showed the need for antivirus programs in the future. It also showed the need to monitor networks and keep them secure. The origins of cybersecurity can be traced all the way back to this very first attack which set the stage for the industry as a whole.

Ransomware is one of the most interesting and also devastating types of cyber attacks. The very first example of ransomware was developed in 1989 and it was known as the AIDS Trojan. Yes, AIDS like the disease. The ransomware was distributed via floppy disks. In fact, attackers would claim that they contained health information or information relating to AIDS research.

When users put the floppy disk into their computers, the AIDS ransomware would proceed to encrypt all of their files, making them inaccessible. Once this was done, it would display a ransomware note on the screen. This note told the victim to send money to a post office box and that if they did so, they would receive the decryption key.

As you can tell it was far less sophisticated than modern ransomware attacks but the overall idea was the same. Rather than using bitcoin like a modern attack, they were forced to tell victims to make payment in cash via the post office. Some victims

actually received a decryption key after making payment but many of them simply never received one or the key that they did receive didn't decrypt the files.

As the payment options in those days were very limited, the attack was mostly localized and affected users in the United States. This was also because it primarily spread via physical hardware in the form of floppy disks.

The AIDS ransomware set a new precedent for holding victims' files hostage in order to extort money from them. Criminals found this type of attack very appealing as they could see the route to direct and significant financial gain.

Other types of attacks that involved disruption or data theft weren't nearly as financially appealing. As the creators were eventually apprehended and faced legal consequences, it showed other potential attackers that cyber crimes could have real world consequences.

It also inspired cyber attackers in other ways. For example, baiting that we spoke about previously using flash drives left lying on the floor takes its inspiration from this very first form of physical media attack. For users, it was one of the first examples of how opening physical media from sources that aren't trustworthy can be dangerous. Not just physical media, but that any unknown files could potentially be harmful.

The next attack that we'll be looking at is the first ever password leak. In 1962 MIT set up the first computer passwords ever. The goal of these was to limit students' time spent on the computers so that they could be shared equally and also to provide privacy. A particular student at MIT by the name of Allan Scherr didn't like this idea.

He decided that he wanted more time, so he created a punch card, which was originally used to program computers, and told it to print all of the passwords in the system. Once these were printed, he used them to get more time on the computers and even distributed them to his friends. One of these passwords was for the teacher's account and they even left them taunting messages.

Allan Scherr actually went on to get his P.H.D in computer science a few years later in 1965. Luckily for us the password leak was the end of his cyber-attacks. He actually went on to have a successful career working for IBM developing operating systems and computer architecture.

Nowadays most cybersecurity attacks use a form of social engineering. One of the oldest and most popular types of social engineering attacks is phishing. The AOL phishing attack that happened in the mid-1990s was one of the first ever instances of phishing and by far the biggest and most successful of its kind at that point.

America Online, commonly known as AOL, was a popular online service provider, before the likes of Google gained popularity. They offered things like internet access, email, and various other services.

The attackers used social engineering tactics and sent fraudulent emails to AOL users. These emails were created to look like they were from AOL and would usually try and convince the users that there had been some kind of a security breach. They would say that their account had been compromised, it required urgent verification, or that they needed to change their password immediately.

When the users clicked on the links provided in the email, they would be taken to fake login pages that closely resembled the official America Online login page. These users would then enter their AOL usernames and passwords onto the fraudulent page, believing it to be legitimate. The attackers would then take these credentials and use them for various malicious purposes.

This attack laid the groundwork for future phishing attacks, and also helped to raise awareness with the public that they shouldn't click suspicious links. It's actually remarkable how similar this attack was to modern phishing attacks today. Despite the fact that cybersecurity has progressed and other types of attacks have become much harder to pull off, phishing has remained almost the same.

This is because rather than exploiting a vulnerability in a system, it exploits the users. Despite all of the advancements in cybersecurity, the average user is still vulnerable to basic social engineering techniques.

While these cybersecurity attacks were often much less complex than the ones today, they were very effective for their time. They laid the groundwork for cyber attacks that would happen many years after them, and for those that we'll continue to see.

That said, they also laid the groundwork for cybersecurity as a field. If those first attacks didn't happen, cybersecurity experts would have been much less prepared for the ones that came later.

Chapter 26: Cybersecurity and The Internet of Things

You understand where cybersecurity came from, what its history is, and some of the first attacks that happened. We've even taken a look at some of the major attacks that happened later and how they shaped the field, forcing more attention to be paid to cybersecurity policies. But what about where it's going? What about cloud storage, artificial intelligence, The Internet of Things, 5G, and biometrics? All of that is a lot to unpack in one chapter, so let's start with The Internet of Things.

The internet of things, commonly shortened to IoT, can be described as devices being connected through software and sensors that have the ability to gather, store, and send data to one another. This is done through the internet and these devices are often connected to the same network in a home or office space. The idea behind this is that by collecting and sharing data with one another, the devices can improve overall efficiency, automate better, and have better decision making.

You can think of this like a town group having a weekly meeting. Individually one of them might know about a problem with a burst water pipe, but together they can make a plan to fix it. One of them might be a plumber, another might have a machine that can be used to dig up the pipe, and others might simply have a free weekend on which they can help. Together, they can solve the problem in the most efficient way possible. This is exactly how the internet of things is designed to work.

Both the term and the concept have been around for at least 25 years, but in the last few years the Internet of Things has truly gained popularity. For example, many people are now wearing smart watches that can obtain information relating to their health like blood pressure and heartrate. This information is then transmitted to a fitness app on their mobile phone. The app might also be linked to their smart scale and have their gym program on it.

This means that the app on their mobile device now knows how much they weigh, how much exercise they're doing, and how their heart is doing. I'm sure you can imagine how useful this could be. Your app can inform you if you need to lose

weight, if your weight is affecting your heart health, and if you're under too much stress.

This same technology is spreading to many areas. It isn't all incredibly complicated, it's often just little changes that can be improved. For example, your light bulb is linked to your alarm clock. Instead of being woken up every morning from a deep sleep, your light bulb can now slowly adjust the light levels until you're gently woken up from a light sleep. This is much less stressful and provides a better start to your day. Your coffee machine can then have your coffee ready ten minutes after that, so that once you're up and eating your breakfast, you can have a nice hot cup of coffee with it.

Or perhaps imagine that you're on your way home from work. You already picked up some milk on the way back because your smart fridge told you that yours is finished. Your thermostat is linked to the GPS on your phone. By the time you open the door your house could already be warm in winter or cool in summer. The lightbulbs know what the time is so if you arrive home after sunset, there's already a light or two on so that you can find your keys easily to unlock the door. Perhaps the door even unlocks itself when it detects your phone in close proximity to the scanner. As you can tell, it's an amazing concept. A life where everything is so easily automated could leave us more time to spend with our families, doing the things we love, and enjoying nature. It will also revolutionize the workplace, allowing for factories to produce things more efficiently and for smart production lines where every machine knows what it should be doing and where the current shortages are.

But it isn't without its challenges and risks. First of all, there are some general technical challenges. What standard do all of the devices use to connect to one another? Each manufacturer designs their devices to communicate with one another but they have no interest in making them compatible with other manufacturers. This means it's easy to get locked into once ecosystem. Perhaps you choose Apple as your device manufacturer of choice. One day you see a great Samsung smart fridge on sale.

You buy it, but now it doesn't communicate with all of your other devices. Certain groups like the EU can develop standards to help combat this but a technical challenge like this will be the least of our worries.

What about adverts? We already see adverts on our computers, our mobile phones, our TVs, on billboards whenever we drive anywhere, and on trains and busses. What if the adverts now spread to our homes? You wake up and your alarm clock tells you why you should buy the matching coffee machine to optimize your mornings. As your shower adjusts to the perfect temperature that it remembers you like, it displays an advert telling you to buy a certain brand of shower gel for better moisturized skin.

Your house stops being a safe place away from the world and becomes filled with adverts, each one telling you to buy some new product that you don't want or can't afford. But what happens when the adverts start to become adware?

If every part of your house is connected to the internet, every one of those connections is a potential security risk. If these devices are all communicating with each other and one out of the fifty that you use has a vulnerability in its software, how long is it before the malware spreads to the rest? Many cybersecurity experts have described the Internet of Things as the biggest risk to cybersecurity in people's future, even bigger than artificial intelligence. Imagine a man-in-the-middle attack where instead of intercepting communication between two people, the attacker intercepts communication between you and your smart devices.

You tell your thermostat that you want it at twenty degrees Celsius and the attacker tells it that you want it at five degrees Celsius instead. This then spreads to a ransomware attack where instead of holding your data ransom, they hold your house ransom. Do you want your thermostat to work properly again? You now have to pay the attackers to regain control of it. What if you don't pay them? Then, they might just decide to turn the temperature on your smart fridge all the way up and allow your food to go off overnight.

You can easily see how every device that's designed to make your life easier could also make it a living nightmare. You might long for the days when your fridge had a simple dial that could be adjusted manually and wasn't connected to the internet.

So, what do you do? Many cybersecurity experts will simply tell you that the best solution is to avoid IoT devices. They're not wrong about this. It's like telling someone that the best way to avoid getting into a motorcycle accident is not to ride a motorcycle. In some cases, it is good advice; not everything is worth the risk that it

poses. But, for some people that isn't a viable option. It's something that they're passionate about, something that they need for their job, or just something that they're not willing to give up.

If the Internet of Things falls into this category for you then I'm not going to tell you to stop using these smart devices. If they make your life better, and easier, then you should continue to enjoy them. But, you should do so carefully. It's important to make sure that your network is secured with a strong password as well as an effective firewall. You also need to choose your IoT devices carefully.

Ideally the ones that you buy should be relatively new so that they will continue to receive updates for the foreseeable future. You should also check which vendors are maintaining support for their devices for the longest. While it might be slightly more expensive to choose these devices, it can also be an investment, as companies that are willing to guarantee longer software updates believe that their devices are going to last a longer time. If that weren't the case, they would have no reason to keep providing updates to them.

The vendors don't just need to support the devices, you also need to make sure that your specific devices are updating. This means having auto updates set to "on" for all of these devices as well as having an internet connection that's capable of supporting all of their update and communication needs. Your smart devices also need to be physically secure. Like with many other types of technology, if an attacker gains physical access to the device, it is only a matter of time before they can gain access to the software and information that's stored on it.

If you're unsure about how to improve the physical security for these devices, pay attention to the chapter that specifically covers physical security. All of the techniques that will work for stationary devices like mobile computers will also work for the physical security on your new smart devices.

You should also have a plan for how you will deal with these devices if there is a security breach. If you're designing an office or home with lots of smart devices it's important that you have a plan in place for how to deal with a potential breach. The plan should ideally focus on first identifying the breach in order to understand what it is.

Next, the compromised devices should be isolated, usually by removing their source of power, so that they cannot affect the rest of the network. Once this is done and the network is secure you can then look at patching the devices in order to fix the vulnerabilities and make sure that you're not a target for repeated attacks.

The Internet of Things is a concept with both amazing potential and massive risks. As it becomes more commonplace to have smart devices in our homes and businesses, you need to weigh the risks for yourself.

Is the added productivity and automation worth the potential loss of productivity and downtime that could occur if there's a security breach? Is the money that you're saving by using these devices greater than the money you're going to have to spend to contain a security breach? Is your data highly sensitive or is it less important and less at risk of being targeted. Only once you've answered all of these questions can you truly determine whether or not the Internet of Things is something that you're ready to adopt.

Chapter 27: Artificial Intelligence and Cybersecurity

In one of our previous chapters, we covered a little bit about the role of artificial intelligence in cybersecurity as part of its history. However, the vast majority of AI's contributions to the field of cybersecurity are yet to come. Many professionals in the IT field believe that it will be the single biggest change in the computing industry as a whole. Cybersecurity professionals are already having to adapt to working with AI and learning how it will change the field that they work in.

At this stage in time, AI is still very much in its infancy. Much of what is marketed as “the newest artificial intelligence technology” is actually just machine learning techniques that we have already been using for years. Many basic algorithms that have been in use for a long time have also been rebranded as artificial intelligence.

This significantly confuses discussion on the field. Because AI has become a buzzword that draws investors and less tech-savvy people in, it's now being used for all kinds of software. It's become a marketing tool used to rebrand old products and software that we previously thought of as simply using code. What does all of this mean?

To understand artificial intelligence's role in cybersecurity we first need to understand what artificial intelligence is. The goal of the field is to develop machines that can “think”, machines that are intelligent and can solve problems. One of the long-term goals of the field is to develop a machine that can perform tasks that a human can. But don't we already have that? Can't ChatGPT write essays like a person could?

This distinction largely comes down to their being different categories of artificial intelligence. At the first level of AI, you have an artificial intelligence that is capable of performing a human task. Not all human tasks or even many, but a single type of task. This is known as Narrow AI or sometimes Weak AI.

An example of this is a computer vision system. These systems take images or videos from the real world examine them in order to produce data. Many phones have

Google Lens and it's freely available on both the Google Play and Apple App stores. You direct your camera at an object and Google Lens attempts to show you information relevant to it.

It does this by looking at things in the image including text. For images without enough information, it might struggle to tell you the exact item. For example, if I point it at the top of my computer mouse it gets the brand right because of the logo but fails to correctly identify the model. If I point it at the underside where the model name is located then it can scan this text and provide me with the exact model and links of where I can buy it.

Another example, perhaps the most famous, is ChatGPT. ChatGPT is an AI based on a large language model. You can ask it questions or give it instructions and it will give you responses in a text format. Users can then respond to it with more instructions, getting it to fine tune or entirely change its responses depending on what they want.

Released at the end of 2022, it quickly became one of the most famous and widely used examples of AI. ChatGPT is largely credited with starting what is known as the "AI boom" as many companies in the wider IT field have now pivoted to using AI. Many people think of it as more "real" AI because it can actually generate content. While it is obviously amazing that AI is being integrated into our daily lives, it's not without its issues.

One major issue users have found with ChatGPT specifically is its "forgetfulness". As users attempt to have longer conversations it will quickly forget the previous prompts. In addition to this many users have noticed that the technology is somehow becoming less effective with time, instead of learning like true intelligence and become more effective.

It can also be inaccurate. If you asked a person in real life an answer to a question that they didn't truly know, they would probably tell you that they're not sure. If you ask ChatGPT something that it doesn't know it will often confidently lie to you. This results in users having to fact check everything and needing to be wary of the content provided by ChatGPT.

Even aside from these issues, ChatGPT is not artificial intelligence in the way that many people think about it. It has no consciousness, self-awareness, or personal

experience. It doesn't actually "understand" the information it generates. It does so based on patterns and knowledge from the data that it was trained on. This means that it is poorly equipped for handling new information. It is only as good as the data it was provided with and it can't judge that some of this data is wrong.

This brings us to the second type of artificial intelligence, AGI or Artificial General Intelligence. This is sometimes referred to as "strong AI" or even "human-level AI". This type of AI can apply its intelligence to a wide range of tasks like a human can and could transfer information from one domain to another. This could feasibly be installed into a robot that could then operate like a human provided it had the physical capabilities. We could use this type of AI to do human jobs and perform general tasks for us. Many people have speculated that the need for humans to work would be erased once this type of AI exists.

Lastly, there is ASI, or Artificial SuperIntelligence. This AI could surpass human intelligence in virtually every aspect. It is theorized that it could solve major human issues with greater efficiency than we ourselves can. ASI, like AGI, is merely a concept at this point in time. As of this point in time there is no "true AI" like you see in science fiction movies where it outthinks humans and takes over the world.

With this in mind it becomes easier to understand where we are with AI and cybersecurity. Specific, narrow, or weak AI, can be developed in order to either launch or protect against cybersecurity attacks. Some parts of cybersecurity are already using AI tools in order to pick up what is important and what isn't so that humans know which parts they need to look at.

Language model artificial intelligence programs can also be used to identify phishing. For example, they could pick up mistakes in emails that phishers try to sneak past people. They can then be used to quarantine these emails or report them to IT staff. At this stage it is not ready to replace actual cybersecurity professionals or the attackers.

It is simply a tool in both of their hands. Naturally, companies running popular AI tools have put safeguards in place to ensure that attackers cannot use their software for malicious purposes.

The problem is, hackers specialize in finding vulnerabilities in systems. This means they're exactly the right people to find vulnerabilities in AI in order to use it for their



own nefarious purposes. Short of that, as AI models become more accessible, it's very likely that people will begin to train them for their own purposes. In the next few years you can expect to see exploits for AI models and probably even guides to training your own "Evil AI" popping up on the dark web.

To summarize all of this, artificial intelligence is here and it's here to stay. Experts have known this since it was founded as an academic discipline in 1956. It's only going to get more advanced with time and as the processing power of computer systems and storage continues to improve.

It's not currently the main tool used for either cybersecurity professionals or cyber criminals but it's only going to continue to grow in popularity with both parties as it grows as a useful tool for humans in general.

Chapter 28: Biometrics in Cybersecurity

Next, let's take a look at biometrics. While people talk about it a lot less than The Internet of Things artificial intelligence, it's having a large impact on the cybersecurity field. Biometrics in cybersecurity refers to the use of unique biological or behavioral characters in order to verify the identity of individuals. Some commonly used types include fingerprint recognition, facial recognition, iris recognition, and voice recognition.

The goal of these is to provide a more accurate form of identity verification than something like a password. Anyone who knows your password can pretend to be you, but if they want to use your fingerprint, they actually need your finger.

Fingerprint verification itself captures and then analyzes the unique patterns of ridges and valleys on a person's fingertips. In recent years this has grown in popularity as a way of unlocking mobile devices, access control for physical workplaces, and verifying who you are when you visit a physical bank branch. Previously, they were primarily used by governments and law enforcements.

Palm recognition works in a very similar way but it verifies a person's palm print. This is used for some access control and authentication but fingerprint verification is usually favored because the devices can be so much smaller.

Iris recognition is a highly accurate method that involves capturing and analyzing the unique patterns in the colored part of a person's eye. These scans are often used for access control systems. At one stage certain smartphones used them but they were removed in favor of smaller bezels.

Another popular method of biometrics is facial recognition. This analyzes features like the distance between a person's eyes, their nose shape, and their jaw shape to identify them. It's commonly used in smartphones but it isn't quite as reliable as the other two. It's been known to sometimes make mistakes.

Less effective versions of facial recognition can be tricked using a picture while others sometimes failed to recognize people after they shaved their beard or when wearing glasses. Many people hope that a combination of iris recognition and facial

recognition will develop in order to make up for the weaknesses in both technologies.

Voice recognition works by identifying what's known as a voiceprint. This is a unique pattern of a person's individual voice such as how high or low pitched it is, the tone, the rhythm, and the pronunciation. This can be used for things like access control and logging in to devices. While this has certain applications, it's known for more false positives than many other types of biometric access.

This is partially because it can easily be impacted by background noises. The best applications for these types of systems are for call centers verifying customers and general voice assistants as well as less secure applications like transcribing notes.

There's also vein recognition which maps the unique vein patterns in a person's hand or finger. This is a newer technology that uses near-infrared light to capture images of the veins. The hemoglobin in the blood actually absorbs this light, making veins appear as darker patterns against a lighter background.

As vein patterns are both unique and stable over time, it's a very accurate form of biometric authentication. Users don't have to physically touch the devices which helps with things like disease spreading. These patterns are also very difficult to replicate, especially because attackers can't even see them. One downside is that the devices themselves are larger than something like a traditional fingerprint scanner.

The last type of biometrics that we'll be looking at is DNA. DNA biometrics analyze an individual's unique genetic code in order to determine who they are. This can be a highly accurate type of biometrics, but it's one that hasn't been used much at all in cybersecurity. This is for a number of reasons.

One consideration is that other things can be determined from DNA like where a person's ancestors came from, who they're related to, or what diseases they might have. There's the ethical consideration that this could be used for something like racial profiling or not promoting a person because they have a chronic illness. For this reason, it's unlikely that we will see cybersecurity use DNA biometrics any time in the near future.

Depending on which method is used, each has their own downsides. In cold countries people have to remove their gloves in order to access their phones when

using fingerprint scanners. The same goes for motorcycle riders and anyone else who works with gloves like healthcare professionals. Iris scanners take up space in a phone's bezels that reduces screen size and facial recognition is less reliable than the other two. Vein recognition is highly accurate but it can take up more space than some other types of authentication and most people don't have the hardware for it at this point in time.

But what do they mean for cybersecurity? Are they a good or a bad thing?

The truth is, they're a bit of both. Let's take a look at why they're better first. As biometrics are unique to each individual, they provide much stronger authentication than something like a username or password. Anyone who sees you type in your password can steal it, but nobody who sees you scan your fingerprint is any closer to being able to unlock your device. They're harder to forge or replicate as copying someone's fingerprint is an incredibly difficult and complex process. Unless you somehow had access to a data leak, you would need physical access to the person to try something like this.

They're also very convenient. As we spoke about in the chapter on passwords, ideally they should be at least 12 characters with a mix of numbers, letters, and special characters. They should also be unique to each account. Biometrics are much quicker; your phone unlocks within a second of you touching your fingerprint to the scanner. You also don't need to remember anything. You can use the same fingerprint for all of your accounts and devices and never have to worry about remembering or saving anything. This also means that they reduce issues like people forgetting about their passwords, people using the same passwords for multiple devices, people using weak passwords, and passwords being vulnerable to brute force attacks.

But it can't all be good. While biometrics have lots of positive attributes, they also have their downsides. The first one is concerns about privacy. In order to use a biometric signature to sign into something, that device or company needs to have your biometrics on record. Whether this is a fingerprint or facial scan, it raises obvious concerns. What happens if an attacker manages to gain access to these files? If they're not properly encrypted, they could find a way to use this biometric data to log in to all of your other accounts. If a password gets stolen then you can change it.

If your fingerprint or retina scan gets stolen you can't exactly get a new finger or eye. Then, there's the ethical and legal implications to think about.

It's important that the data is properly protected to stop unauthorized access. One also has to consider what attackers could do if they gained access to certain biometric information. In the future could they plant fingerprints at crime scenes using your unique fingerprint biometrics? While this may not be an issue now, technology moves incredibly fast and it could be an issue in the near future.

There's also the issue of standardization. Many of the companies seem to have their own unique biometric technologies, and none of them want to agree on specific standards. This means that if you change devices or systems, you can't simply copy your biometric information across. Lastly, there's the issue of false positives and negatives. Not all biometric systems are equally accurate and if they're being used as the sole method of authentication, it can cause issues. Fingerprint scanners don't work if your fingers are too wet, facial recognition struggles with glasses and beards, and voice recognition just isn't accurate enough altogether.

A large part of cybersecurity is balancing risks. Security features are meaningless if users turn them off because they find them to be too much trouble. Biometrics might not always be better than a strong 12-digit password made using completely random letters. But they're definitely better than a 4-digit pin or a password like "1245678".

They're also generally much faster than passwords which means that they can be added as an additional layer of security on top of a password. An account using both a password and biometric access will be very difficult for attackers to gain access to. It's up to the companies who hold users' biometric data to ensure that this data can't easily be obtained by attackers.

Overall biometrics represent progress in the cybersecurity industry. They're difficult to copy and add an extra layer of security to certain accounts. As more devices are shipped with biometric scanners, biometric access will quickly become one of the easiest and most commonly used methods of two factor authentication.

Chapter 29: Other Emerging Trends in Cybersecurity

So, we've looked at three of the main emerging trends in cybersecurity, namely The Internet of Things, and artificial intelligence. But what about all of the others? What about 5G, biometrics, cloud security, and all of the other emerging trends? Don't worry, that's what we'll cover in this chapter! If the future of cybersecurity was a burger, artificial intelligence would be the patty, The Internet of Things would be the bun, and biometrics would be the lettuce.

But who wants to eat a dry burger without butter, sauce, or any of the other delicious ingredients? Some of these trends might gain less headlines than AI and the IoT but they're still highly important parts of where the cybersecurity field is going.

Let's start with 5G. I'm sure you've heard a bit about it, but I'm not sure how much of what you heard was accurate. There have been conspiracy theories and wild claims floating around ever since the technology was announced. People have claimed that it can control the weather, that it was responsible for the Covid-19 pandemic, that it causes cancer, and that it's used for mind control. It's safe to say that none of these theories have any scientific backing or real merit. So, what is 5G?

It's actually quite simply, 5G is the successor to 4G. It's the latest standard in mobile telecommunications. It's also managed to bring about several improvements over 4G in terms of speed, latency, and general connectivity. 5G uses different radio waves to 4G and in doing so it can achieve much higher speeds. But that comes at a cost. It also has a much shorter range, which means that it's only going to be rolled out in densely populated areas for now.

With any new technology there's always going to be some security challenges. Manufacturers try to ensure that devices are as secure as possible on release but it often takes a few updates and security patches before they find and eliminate all of the major vulnerabilities. 5G is however designed based on the previous technologies of 4G and before that 3G for which providers had many years to improve on security

protocols. What does all of this mean? It means that on release and for the first few years you can expect 5G to be slightly less secure as exploits continue to be found.

However, given time it will become more secure than any of the previous technologies. 5G comes with improved authentication protocols for both users and devices as well as end-to-end encryption which is both effective and secure. Overall, the technology provides less new security risks than many other developments and perhaps its biggest impact on cybersecurity will be the increased internet speeds. These speeds will make other technology like the Internet of Things more widespread and viable, leading to its own cybersecurity challenges.

Next, let's look at the cloud storage. With cloud storage, data is stored remotely in other places and then is available via the internet. As we spoke about earlier, it can be very useful. Backing your data up to the cloud means that if you suffer a ransomware attack, you can still easily recover all of your important information. As the internet becomes faster with technologies like 5G being rolled out, it becomes more viable to use cloud storage for data that is regularly used. Economies of scale also dictate that it can be cheaper to have less storage on a specific device and more data stored in the cloud.

For example, instead of keeping all of your pictures and videos on your mobile phone, you keep most of them on the cloud. This frees up plenty of space for other applications, even as pictures and videos increase in size. If you want to show someone an older picture you can still access it, your phone just quickly fetches it from the cloud storage. Devices have also grown smarter which means that they can automatically detect what data needs to permanently be available offline and what data only needs to be stored in the cloud. This is incredibly convenient to use.

Another advantage is that you can access the same information across multiple devices. Imagine that you are writing a book on your laptop. You're at work and you want to show one of your colleagues your book while you're having lunch but you don't bring your personal laptop to work. That's okay, in a matter of seconds you can have it on your mobile phone.

There are also downsides to this technology. If you want to access a specific file that's stored on the cloud and you're in an area with a slow internet connection, you could not be able to access it or have to wait a long time. Then there's also the data cost.

Depending on where you live in the world, data could be incredibly cheap to the point of almost being free, or it could be very costly.

Constantly uploading and downloading files can end up costing much more in data than it's saving in physical storage costs. There's also the problem that this storage isn't bought, it's rented. While it might be cheaper in the short term, sometimes it can add up if you choose to store large amounts of data on the cloud for years and years. But these are mainly technical challenges. What about the cybersecurity challenges?

The first major issue with cloud storage from a cybersecurity point of view is the fact that it's hosted online. All someone needs to access all of your private data is your password. There's the risk that people won't use secure passwords, or will reuse passwords that are originally for other accounts. There's also the risk of social engineering, like we spoke about with celebrity iCloud accounts and how they fell victim to having their private pictures leaked. This is the most likely way that an attacker would gain access and it's happened before.

Aside from this there is the fact that there could potentially be a security breach within the company itself. If you use Google to store your data in the cloud and they have some kind of security breach, all of your data could be a part of this breach. While some experts would argue that this is less of an issue with large companies like Google and Apple that have very advanced cybersecurity procedures, it wouldn't be the first time a company this large suffered a breach.

After all, Yahoo was one of the biggest online service companies in the world and they had a massive cybersecurity breach that wasn't even disclosed to the public until years later, when their information was already being sold on the dark web.

If information is stored online, it's always going to be easier for attackers to access it than if it's stored offline on a hard drive that's locked in a safe. This is simply a risk of using this technology. As cloud storage continues to grow in popularity it's going to be increasingly important that you choose a provider that values security and also that you use very secure passwords for these accounts.

Another emerging cybersecurity trend is the increased use of automation. Most monitoring and incident response was, and often still is, typically carried out by individuals. People were paid to monitor network traffic and system logs. If they

noticed anything suspicious, they would investigate it further or report it to someone who had more knowledge on the matter. Now, it's increasingly common for this to be automated.

Automated tools are used to continuously monitor network traffic, system logs, and other data sources for any signs of malicious activity. This is done with machine learning algorithms that can analyze patterns and anomalies in order to more quickly identify potential threats. The incident response is then also automated. There are now incident response tools which can perform tasks like isolating affected systems, blocking malicious IP addresses, and initiating predefined response plans. All of this is done without any need for human intervention.

This software is also increasingly being used for managing software updates and patches. If this is automated there's no way that anyone forgets about it or the person whose job it is to do the patches happens to go on sick leave. As you remember from the chapter on previous attacks, the infamous WannaCry ransomware attack was only an issue because people were running out of date versions of Windows and had not installed the latest security patches.

This type of automation goes hand in hand with artificial intelligence. There is in fact significant overlap between the two fields, as many automation tools are incorrectly labelled as AI in order to boost sales and investor confidence. As automation gets more advanced, you can expect more and more parts of cybersecurity to be automated. It's highly likely that in a few years' time almost no users will actually check their own software and manually approve updates. It will all be done automatically by an update manager and the end user will never even see any of this happening.

The last emerging trend that we'll be taking a look at is something that's known as a Zero Trust security model. Traditionally, entities within a network perimeter were trusted by default. This sometimes resulted in vulnerabilities as attackers could impersonate a trusted device.

Alternatively, if they gained access to the network, they were then one of the trusted devices. Once they had gained access to the network, they could essentially gain access to anything inside it. The Zero Trust security model instead operates on a "trust nobody, verify everything" approach.

This approach is obviously more secure but it's also more resource intensive. As this continuous verification of the identity and security of network users is happening, there's a lot of authentication and authorization checks being done. Users are also granted the lowest level of privileges that they need to perform their tasks.

If they don't need access to the internet, they aren't given it. If they don't need access to a specific file or portion of the network, they aren't given it. Whilst some users will naturally find this painful, it's a much more secure way of operating.

The access and privileges that users have is also fluid. It's constantly changing depending on a variety of factors. For example, they're physically at the office they might have access to more data than if they're working from home. If they're using a different device than they normally do, say a personal phone because their work one hasn't arrived yet, they will also have a lower level of access.

Zero Trust also incorporates things like multi factor authentication instead of just standard passwords and a variety of other best practice cybersecurity techniques. As cybersecurity becomes more important and devices become more advanced, it's likely that Zero Trust will continue to gain traction as a cybersecurity model.

Cybersecurity is a field that moves incredibly fast and it's possible that these emerging trends will be entirely adopted, or perhaps entirely obsolete in just a few years' time. How fast it moves is part of what makes it so interesting, and I'd encourage you to take a look at some of the latest trends and see if they still line up with what we've covered or if there's anything new that you find interesting.

Chapter 30: Is Cybersecurity the Right Career For You?

We're almost at the end of this book and if you weren't considering a career in cybersecurity when you started it, there's a decent chance you are now. And why wouldn't you be? We've already established various great things about the field. Let's look at a few of them:

As people use more computer products in their daily lives, the number of potential vulnerabilities and cyber threats continues to increase. Cybersecurity professionals play a vital role in ensuring that systems, networks, and data are protected from a wide range of attacks.

What does this mean for the field?

It means that the demand for highly skilled professionals already is, and will continue to increase as time goes on. Even now the demand for cybersecurity professionals far exceeds the supply. This means that organizations are actively looking for qualified people who can improve their cybersecurity systems and procedures. Having a job look for you instead of having to look for a job certainly sounds appealing.

This goes hand in hand with job security. Because people with these skills are always in demand, there isn't an oversupply of labor for the positions. You won't constantly have to compete with young people looking to push you out of your job, because there are plenty of jobs for them too. This means long-term job security for you and also the availability of jobs anywhere that you want to work. The field exists everywhere in the world with computers and that means if you want to immigrate or work overseas for a few years, it's relatively easy to find something.

While salaries differ depending on where you live and what your level of experience is, salaries in cybersecurity are generally rather appealing. This is largely because of basic supply and demand. There are more companies looking for cybersecurity professionals than there are people willing to fill those roles. Because of this companies have to increase their pay and benefits in order to try and attract the best

people to them. It's not a field where they can simply hire an underqualified person and hope everything works out okay. They need the best of the best to ensure that their networks and data are safe from potential attackers.

Cybersecurity is also a rewarding field to be in on a personal level. The work that you're doing is impactful in the best kind of way. You're making sure that people and companies stay safe from potential attackers. Knowing that you saved somebody's files from being deleted by ransomware or stopped their private data from being leaked online can be very rewarding.

It's also a field known for collaboration and ethical standards. Because it deals so closely with criminals, the field is governed by strong ethical standards and legal boundaries. If a new virus is released that's damaging systems, these professionals often work together to figure out ways to stop it and remove it from people's computers or networks as quickly as possible. There are very few fields with this kind of collaboration between countries and companies. Not many fields have so many people with the exact same goal who are willing to work together to achieve it.

This is also a field with a wide array of roles. From ethical hackers and penetration testers to data analysts and security consultants, there are a lot of available positions. Each of these positions has its own unique benefits and characteristics. You could choose to specialize in network security, system security, or perhaps cloud security. If you're interested in what the specific positions are and what makes them different, make sure to keep going. We'll look at the main ones soon and hopefully give you a better idea of which one could be for you!

But are you the right kind of person for cybersecurity? It's by no means an easy field and there's some important things that you need to make sure of before you get into it.

The first and possibly most important thing is that you're someone with an ethical mindset and strong morals. Learning the same skills as criminals who are out to make as much money as possible can be very tempting. You'll also be in a position where you have control over the safety and security of a lot of people and their data. It's important that you prioritize legality, integrity, and confidentiality when working.

It's also important for cybersecurity professionals to be curious and have good problem-solving skills. It's a field in which you will need to figure things out on your own. In order to find vulnerabilities in a system, you must first understand it. If you're someone who like to figure out how something works, or who looks at something and sees places where they can make improvements, then you have just the right mindset for the job.

Another crucial part of being a cybersecurity professional is that you're adaptable and willing to commit to continuously learning. Like many other IT fields, it's a landscape that's constantly changing and evolving. Cybersecurity is credited as beginning as a field in 1971. This means that all of its developments up until now have happened in a single lifetime. From the first virus, to the first antivirus, to the first DDoS attack, all of it has happened very recently in the greater scheme of things. This means that the most important thing isn't knowing about one specific vulnerability or type of attack. It's more important to be able to constantly learn and adapt and to have a base understanding that can be built upon.

This also makes communication and collaboration skills incredibly important. You need to be able to understand new threats, but also to be able to explain them to the people you help protect and to the people you work with. Complex threats require teams of people to understand and combat and if you can't work with others then you can't be a part of these teams.

The better you are at understanding other people and explaining yourself to them, the better you can work with these people to solve problems and keep the rest of the world safe from attacks.

You also need the "hard" or technical skills required for the field. To understand vulnerabilities and exploits you first need to understand the systems. And you cannot understand the systems unless you've studied how they work. It's important to have a strong grasp of computer networking, operating systems, and various programming languages.

If you want to specialize in a part of the field then other specific technical skills like network skills or cryptography can also be incredibly beneficial. I'm sure you're beginning to understand that this is by no means an easy field. If you're simply

looking for the easiest way to make a bit of money there are other fields that you would be better off in.

The time needed to learn the systems, before you can even learn how to protect them, is significant. As you'll see when we look at the individual positions within the field and what you need to study for them, it's a long road to being a true professional.

This is why this next point is so important. To become a cybersecurity professional, you must be truly passionate about computers and the field of cybersecurity. It's going to take a long time to learn all of the required skills and if you aren't truly passionate there's a good chance that you'll give up halfway.

You need to be someone who really cares about ethical behavior, securing digital systems, and keeping people's information safe.

If you have most or all of these qualities, I'm confident that it will be a field that you find enjoyable and genuinely rewarding, both on a financial and emotional level.

Chapter 31: What Roles Are There and Where Do You Study them?

By this point you should know whether or not you have the required qualities to be a cybersecurity professional. Hopefully you also have some idea of whether or not this is a field that you want to get further into and learn more about. Now that you know these things, it's time to look at some roles in the field. Because this book is based on the fundamentals of cybersecurity, we'll start with some of the simpler roles that require less studying and technical knowledge.

As we move further down the list, we'll get to the roles that require more extensive experience as well as advanced technical knowledge and years of studying. It's important to understand that cybersecurity itself is quite an advanced field to work in. Most cybersecurity professionals start in general IT before graduating to working in security.

The first role we'll be looking at is a security analyst. Security analysts are generally responsible for monitoring more than actively protecting. A large part of their time will be spent monitoring various tools and systems that the company uses in order to identify potential security breaches or intrusions. This is commonly referred to as SIEM, which is short for Security Information and Event Management. This is usually done in real-time as the applications and network hardware generate security alerts.

It's important that these incidents are dealt with promptly when detected. This means that the security analysts need to look at the logs, decide how serious it is, and then take action. Often with more serious threats they will flag it so that someone more experience and/or technically proficient can take a look at it. The security analyst will develop a report on the incident as well as the vulnerabilities and overall security.

This report will then be looked at by people in higher up positions to determine what needs to be done to improve security or mitigate the threat.

Another important part of the role is education. As security analysts are lower down than the other positions, they're more likely to be tasked with interacting with end users. They might teach them about best practices for security like having strong passwords and how to recognize fake emails and other phishing attempts. Much like you after reading this book, they know more than almost all end users, and they try to share some of this information to keep them safe.

Security analysts need to have strong analytical and problem-solving skills in order to analyze the data and understand how to solve the potential incidents. They also need good communication skills for their reports as well as a basic understanding of operating systems and networking. Lastly, they need to be familiar with the security tools used.

They often have a bachelor's degree in computer science, programming, or something similar. However, employers are more focused on the skills than the qualifications due to shortages in the field.

The next role that we'll be looking at is that of a network administrator with a focus on security. Where the security analysts' job is mostly to observe, the network administrators are more likely to be actively implementing systems. One of their main jobs is to configure and maintain the infrastructure used for network security. This includes firewalls, setting up VPNs (virtual private networks), and taking care of other security devices.

They will also usually be responsible for developing, implementing, and enforcing security policies for the network. This involves things like making sure that only authorized people can access the network, deciding which sites need to be blocked or allowed on the network, and establishing guidelines so that users of the network know what they are and aren't allowed to do. Having well thought out and implemented policies can save a lot of time and money further down the line.

The network administrator who is focused on security is also likely to troubleshoot security issues with the network. This could involve investigated users trying to access unauthorized sites and services but also addressing other vulnerabilities. They are also responsible for investigating network anomalies and generally ensuring that the network remains safe and operational.

So, what do they need to be able to do all of this?

Firstly, they need a strong understanding of networking administration. This includes things like configuring routers, switches, and access points. They also need to understand different protocols like TCP/IP, which is a set of rules that determines how a computer connects to the internet.

They must have a basic understanding of cybersecurity principles like encryption, access control, separation of privileges, and user privacy. In addition to this it's important that they are familiar with firewalls as a part of networking and can configure and manage them.

Network administrators who are focused on security also need to stay updated on security threats and vulnerabilities as it's a constantly changing industry. Lastly, most places require a relevant certification, for example Cisco's CCNA Security or CompTIA's Security+ certification. In addition to this most employers are looking for candidates with a bachelor's degree in information technology, network engineering and security, or something else similar.

This position is a good stepping stone for cybersecurity professionals for who want to progress to more advanced positions as they grow in their career. The network knowledge and experience dealing with threats gained in this role can be vital for more advanced ones.

The next role we'll be looking at is that of incident responder.

Incident responders focus on specific security events that occur. They will investigate these events and then take the necessary actions to ensure that the security breach or other issue is taken care of. They are also responsible for developing incident response plans that outline how to respond to certain threats in order to best mitigate them.

It's important that they're familiar with forensic tools used to understand the scope and severity of any breach in security. These tools can also be used to collect evidence from the attacks that can either be used to identify the attackers or to develop ways to prevent similar attacks in the future. They need to have strong communication skills as they are the professionals who often have to deal with people affected by the security threats, while also communicating the technical information required to more advanced members of the security team.

Employers are generally looking for candidates with a degree in computer science, electrical engineering, information assurance, or cybersecurity specifically. Security certificates like a certified information security manager certificate (CISM) or incident response-specific certifications are also highly beneficial.

Next are cybersecurity consultants. Like other type of consultants, they get to move around, giving advice to different departments or businesses on how to improve their security systems and handle threats. It's a very interesting job as you can travel the world, learning about different types of businesses and visiting different countries, all while getting paid.

Cybersecurity consultants do however have to have an advanced understanding of multiple different parts of cybersecurity. They need to do risk and security management assessments of the company, to see where the likely security breaches would occur. They then need to figure out the best strategy to mitigate these risks while also saving the client as much money as possible.

It's common for cybersecurity consultants to be hired to develop security policies and procedures for the company. These guidelines are then used by lower-level IT staff working for the companies in order to ensure that things like access control and incident response are handled correctly. As they're interacting directly with clients and staff at companies it's also very important that they have excellent communication skills. This is both for writing the guidelines and reports and also for speaking directly with the staff.

Another reason they need good communication skills is that they are also often hired to provide security training to staff. People take it more seriously when a qualified professional tells them about best security practices, what threats there might be to the company, and how to recognize and respond to potential security incidents.

As you can see, cybersecurity consultants need in-depth knowledge of the principles, technology, and best practices associated with the industry. They also need to be as up to date as possible, because systems that they put in place might need to continue working for the companies for years.

Similarly to the other positions, it's important that they have a degree in computer science, information security, cybersecurity, or another similar field. Additional certificates like a CISM (Certified Information Security Manager) and CISSP

(Certified Information Systems Security Professional) can also help to ensure that candidates are adequately prepared for the role.

One cybersecurity role that many people aspire to is that of a penetration tester, otherwise known as an ethical or white-hat hacker. Ethical hackers need to think like any other type of hacker would. They look for vulnerabilities in systems, networks, and applications.

They get to simulate real-world attacks in order to see where these vulnerabilities are and what needs to be improved in a company's systems. This means gaining access to things that they theoretically shouldn't be able to as well as seeing how the target deals with the attack.

Once they've done these tests, they need to document their findings in a report. These reports will have what methods they used, how they worked, and how to fix the vulnerabilities that they identified. Once this is done, they will usually work closely with the security teams in order to make sure that they understand what needs to be done to fix the vulnerabilities and that it's being done correctly.

Penetration testers need to be up to date on the latest techniques and exploits in order to ensure that they can accurately check the vulnerabilities that certain systems might have. You can expect them to know about the best ways to carry out a phishing attack or the most effective ways to crack users' passwords.

Penetration testing jobs usually require a bachelor's or even master's degree in cybersecurity, computer science, or IT. Similarly to the other positions, additional certifications in cybersecurity are also beneficial. In addition to this there are some penetration tester specific certifications.

The most senior position that people will usually reach in cybersecurity is that of CISO, or Chief Information Security Officer. As the most senior person responsible for the cybersecurity department, the Chief Information Security Officer is responsible for overseeing all of the departments. This includes penetration testing, incident response and crisis management, risk management, and managing the cybersecurity budget. The CISO will usually report directly to the CEO of the company.

It's their responsibility to make sure that all of the departments are running smoothly and that threats are dealt with quickly and efficiently. Because they need to

deal with so many different departments, it's important that they have experience in a variety of cybersecurity positions before they rise to Chief Information Security Officer. They also need strong leadership skills, as well as effective communication skills in order to deal with the top leadership of the company and the cybersecurity employees working under them.

It's typical for a CISO to have a master's or even a PhD in cybersecurity, or information technology. Certifications such as a Certified Information Systems Security Professional (CISSP) or a Certified Chief Information Security Officer (CCISO) can also be beneficial. This is the pinnacle of cybersecurity positions and something that many people spend years trying to achieve.

We only looked at some of the popular positions in the field of cybersecurity but they should give you an idea of what you can work towards and what kinds of degrees or certifications you need.

Chapter 32: The Route to Cybersecurity

As you could tell from the previous chapter, none of the cybersecurity positions are truly entry level. They mostly require extensive studying on top of certifications. Lots of them also require years of working in the IT field in order to reach that level. For this reason, we're going to be looking at some of the other positions you can get into. These positions aren't directly in cybersecurity but they can put you on the right path to reaching it. They provide strong foundations in the required skills and also help to open doors into the field of cybersecurity. We'll also look at the best degrees to study in order to reach the ultimate goal of becoming a cybersecurity professional.

If being a cybersecurity professional is something that you truly want, it's important that you don't give up or feel discouraged by the high entry requirements. The other IT jobs that you might work in while you gain the necessary experience are often very rewarding themselves, both in terms of job satisfaction and in terms of financial remuneration. But enough of that, it's time to look at some of these rewarding IT positions.

The easiest way to start your journey to being a cybersecurity professional without having a degree or advanced certification is to start working in IT support or the help desk. This helps to give you a foundational understanding of IT systems, troubleshooting, and user support. It especially shows future employers that you have the necessary communication and people skills to work in other more advanced positions. The same goes for a physical IT support specialist. From these roles it's easier to move to other more advanced ones like network administration, system administration, or security analysis.

This brings us to our next role, that of a network administrator. Standard network administrators are less focused on cybersecurity than those we spoke about in the previous chapter. Nevertheless, it's an easy and logical career progression to move from one to the other. A standard role in network administration will help to give you experience with network security principles, firewalls, and other important cybersecurity fundamentals.

Quality assurance testing is a job that not many people realize can help get you into cybersecurity. The role of these testers is to check software to ensure that it functions correctly and meets specific requirements. If it doesn't, they'll then report these bugs so that they can be fixed before release. While there are some flaws that only come out once something is released to the public, they help to ensure that most releases are significantly more stable and bug free. One of the major requirements of new software is that it's secure.

Part of quality assurance testing involves testing for vulnerabilities in new software. This is almost like a very basic form of penetration testing, and it develops a good understanding of security standards as well as a general understanding of cybersecurity. This can be very useful when transitioning to a career in cybersecurity.

Any kind of work with databases and generally handling user data can also be a good way to start out a career in cybersecurity. Data entry specialists and database administrators both have to deal with various forms of data management. This data is often sensitive and the roles require a good understanding of the requirements around user privacy, handling user data, and cybersecurity data policies. While it's quite a big jump to go from that to general cybersecurity, companies that deal with databases usually have cybersecurity professionals working for them as well as database entry specialists and administrators. T

his means that you can sometimes work your way onto a cybersecurity team within the same company or start doing more basic work for them in your current role.

System administration is also a good stepping stone to cybersecurity. It focuses on the management and maintenance of IT infrastructure which includes servers and other systems. Much like network administration, there's plenty of relevant cybersecurity experience to be gained, as you're sure to have exposure to security practices and configurations. You might even deal with cybersecurity professionals if the company hires a consultant to help put certain systems in place or advise on how to improve your security.

Junior IT auditors can also work their way to careers in cybersecurity. This role is quite well suited for someone who wants to be a cybersecurity analyst or consultant in the future. A Junior IT Auditor generally helps more senior auditors to check the effectiveness of IT controls, security measures, and standard compliance. This



means that as a Junior IT Auditor you'll develop a good understanding of cybersecurity standards, how to evaluate a company's security, and what IT controls are most effective for ensuring better cybersecurity.

You'll also be exposed to a variety of different companies so you'll get to see how their IT departments function and what kind of cybersecurity controls they use. If you have or are planning to study a qualification in information technology or computer information systems, this job is the perfect way to get into a cybersecurity career.

Next is the role of IT analyst. IT analysts are generally looking for ways to improve a company's IT systems. They need to look at different parts of the overall systems and then figure out what needs to be done to keep them updated. This also extends to other things like troubleshooting, and making sure that best practices are being used.

Being able to work with so many different parts of a company's cybersecurity is what makes this such a good transition. An IT analyst already has analytical skills, an understanding of IT systems, and a good familiarity with networking concepts. All of these skills help to make the transition into a cybersecurity career easier.

If you're in a position where you can afford to not earn a lot for the first while, it's also a good idea to consider an internship. Many IT departments offer internships, especially for young students. This can be a good way to learn more about general IT which will help with cybersecurity. It can also be a great way to meet people.

A lot of finding a job is about who you know and if you meet people in a cybersecurity department, they can be invaluable when it comes to finding a job later. When they need a new person in their team, they're likely to remember you if you made an impression. This gives you a significant edge over other candidates when applying for a position.

If you're already in one of these positions, the best thing that you can do is start to learn cybersecurity while working. If you can start to help more with the cybersecurity tasks while working in your current role, it will be highly beneficial. If you can't do that at the company you're working at, you can always spend some of your free time learning about cybersecurity.

Conclusion

Congratulations on making it to the end of the book! If you made it this far then you've learnt a lot about cybersecurity. You know who the attackers are, who protects against them, what types of attacks there are, and how they're carried out. You've learnt about some of the most important cyber attacks in history, ones that truly shaped the industry into what it is today.

You also know how to better protect yourself against them whether it be your business, your computer, your mobile device, or your online accounts. If, or more probably when, it does happen you know what to do. How to recover from an attack, and how to make sure one never happens again.

You now know what the emerging trends are in cybersecurity, and how things like artificial intelligence, the internet of things, 5G, and biometrics will continue to shape the field. While nobody can perfectly predict the future, you should have a reasonably good idea of where it's going to go in the next few years.

You've even learned about how to turn cybersecurity into a career. It's a long journey but a rewarding one and you know which qualifications and jobs you need to get you there. By now should also know whether or not it's a field that you actively want to pursue as a career or just something that you enjoy learning about and find interesting.

Maybe you just needed to figure out how to improve the cybersecurity in your small business without having to pay for a professional. Whatever your reason for reading it, this book wasn't just a collection of facts. It was a call to action. To improve your cybersecurity and the cybersecurity of those around you.

Cybersecurity is a constantly evolving field and in just a few years' time there could be new attacks, new massive breaches to learn about, and new technology that comes with its own set of vulnerabilities. It is a field with a huge amount to learn, and this book only scratches the surface on some of the more complex concepts and attacks.



There's still so much more for you to explore about how the code works, how each individual attack functions, and how it's going to integrate with artificial intelligence in the future.

You did a great job by finishing this book, but I'd encourage you not to leave it here. You should always keep learning, keep growing, and keep improving your knowledge.

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